

Habitat fragmentation mediates the mechanisms underlying long-term climate-driven thermophilization in birds

Reviewed Preprint

v2 • October 30, 2024

Revised by authors

Reviewed Preprint

v1 • July 12, 2024

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eLife Assessment

This **fundamental** study substantially advances our understanding of how habitat fragmentation and climate change jointly influence bird community thermophilization in a fragmented island system. The authors provide **convincing** evidence using appropriate and validated methodologies to examine how island area and isolation affect the colonization of warm-adapted species and the extinction of cold-adapted species. While minor clarifications regarding the definition of fragmentation could further enhance the presentation, the study is of high interest to ecologists and conservation biologists, as it provides insight into how ecosystems and communities respond to climate change.

<https://doi.org/10.7554/eLife.98056.2.sa2>

Abstract

Climatic warming can shift community composition driven by the colonization-extinction dynamics of species with different thermal preferences; but simultaneously, habitat fragmentation can mediate species' responses to warming. As this potential interactive effect has proven difficult to test empirically, we collected data on birds over 10 years of climate warming in a reservoir subtropical island system that was formed 65 years ago. We investigated how the mechanisms underlying climate-driven directional change in community composition were mediated by habitat fragmentation. We found thermophilization driven by increasing warm-adapted species and decreasing cold-adapted species in terms of trends in colonization rate, extinction rate, occupancy rate and population size. Critically, colonization rates of warm-adapted species increased faster temporally on smaller or less isolated islands; cold-adapted species generally were lost more quickly temporally on closer islands. This provides support for dispersal limitation and microclimate buffering as primary proxies by which habitat fragmentation mediates species range shift.

Overall, this study advances our understanding of biodiversity responses to interacting global change drivers.

1 Introduction

One widespread consequence of anthropogenic-caused climate warming is the shift in species' distributions toward cooler regions (Chen et al., 2011 [DOI](#); La Sorte & Thompson, 2007 [DOI](#); Thomas & Lennon, 1999 [DOI](#)). As species move across the globe, the composition of community changes, with species both colonizing and departing (Menéndez et al., 2006 [DOI](#); Tingley & Beissinger, 2013 [DOI](#)); yet colonizing and departing species are unlikely to be randomly drawn from species pools, but instead will be drawn depending on their traits, such as being warm-adapted versus cold-adapted. As a result, a warming climate should lead to compositional shifts toward relatively more warm-adapted species in local communities (Thomas et al., 2006 [DOI](#)), a phenomenon termed “thermophilization” (Feeley et al., 2020 [DOI](#); Gottfried et al., 2012 [DOI](#); Zellweger et al., 2020 [DOI](#)).

In practice, thermophilization can be measured by a temporal increase in the Community Temperature Index (CTI) (Devictor et al., 2008 [DOI](#)) – an estimation of the average temperature experienced by all species in an assemblage throughout their respective ranges. While thermophilization of communities is usually accompanied by an increase in warm-adapted species and a decrease in cold-adapted species (Curley et al., 2022 [DOI](#); Prince & Zuckerberg, 2015 [DOI](#); Tayleur et al., 2016 [DOI](#)) (Figure 1a [DOI](#)), this process is usually spatially heterogeneous and depends on the availability of current habitat and the ability to disperse into newly suitable habitats (Gaüzère et al., 2017 [DOI](#); Richard et al., 2021 [DOI](#)), both of which will be directly or indirectly affected by other global change stressors. Habitat fragmentation, usually defined as the shifts of continuous habitat into spatially isolated and small patches (Fahrig, 2003 [DOI](#)), in particular, has been hypothesized to have interactive effects with climate change on community dynamics. Specifically, habitat fragmentation usually alters both habitat spatial structure and microclimate characteristics (Ewers & Banks-Leite, 2013 [DOI](#); Fahrig, 2003 [DOI](#)), either of which can influence thermophilization processes (Gaüzère et al., 2017 [DOI](#); Platts et al., 2019 [DOI](#); Richard et al., 2021 [DOI](#); Zellweger et al., 2020 [DOI](#)). Understanding the interaction between climate change and habitat fragmentation on community composition is essential to predicting biodiversity outcomes but we generally lack empirical examples with which to study this phenomenon (Opdam & Wascher, 2004 [DOI](#)).

Theoretically, habitat fragmentation could mediate differential dynamics of warm-adapted versus cold-adapted species either by changing existing habitats' suitability or by limiting dispersal into newly suitable habitats (Newson et al., 2014 [DOI](#); Opdam & Wascher, 2004 [DOI](#)) (Figure 1 [DOI](#)). The overall effect of fragmentation on climatic niche tracking of communities can be decomposed into three different mechanisms, each of which has been proposed and supported with different degrees of evidence. First, fragmented habitats may more weakly buffer macroclimate than intact habitats (Ewers & Banks-Leite, 2013 [DOI](#)), resulting in fragmented patches experiencing relatively higher temperatures. Increased temperature, especially in small patches, may boost the colonization or population growth rate of warm-adapted species but cause higher extinction risk or demographic declines of cold-adapted species (Figure 1b [DOI](#)). This microclimate mechanism has been proposed as an important factor driving thermophilization in plant communities (Zellweger et al., 2020 [DOI](#)).

Second, larger patches generally contain higher habitat heterogeneity and therefore also encompass higher food availability and microrefugia (Liu et al., 2020 [DOI](#)). A diversity of habitats and resources is closely related to a species' successful establishment and can act as a buffer against extinction under climate change (Luoto & Heikkinen, 2008 [DOI](#)). For instance, higher habitat diversity can boost the ability of niche tracking in bird communities (Gaüzère et al., 2017 [DOI](#)), mainly by fostering species' dispersal into new, climatically suitable sites. Similarly, in areas where topography generates greater microclimate variation, population losses of plants and insects due

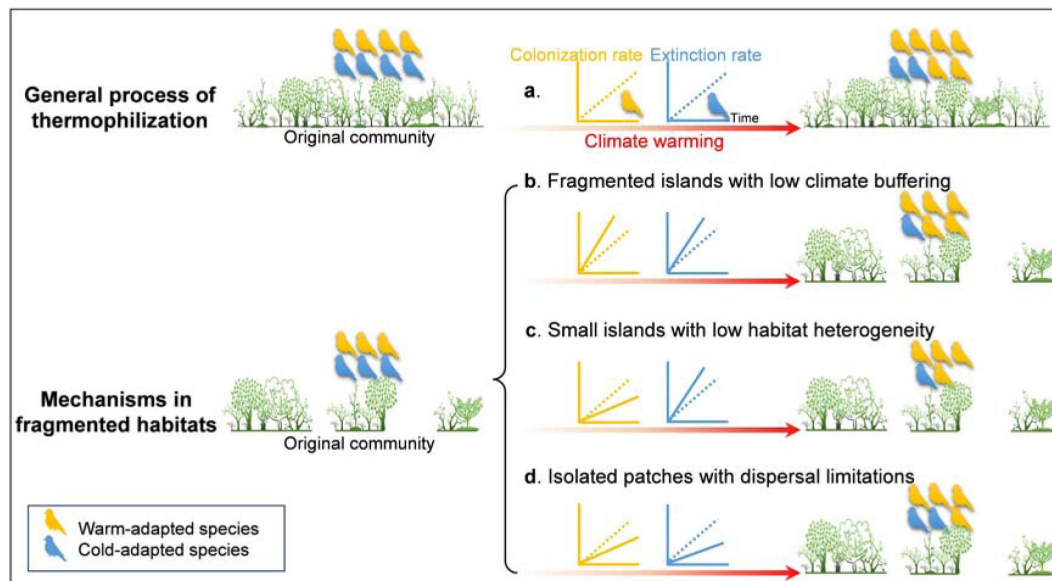


Figure 1.

The general process of thermophilization and the hypothesized framework of thermophilization in fragmented habitats.

Two general processes of thermophilization are shown on the top: increasing colonization rate of warm-adapted species, and increasing extinction rate of cold-adapted species over time (a). Compared to continuous habitats, a hotter microclimate caused by lower buffering ability on fragmented patches may attract more warm-adapted species to colonize while causing cold-adapted species to extirpate or emigrate faster (b). Loss of cold-adapted species can also be exacerbated when highly fragmented patches harbor lower habitat heterogeneity (e.g. resource, microrefugia), which will also reduce colonization of warm-adapted species (c). Isolated patches due to habitat fragmentation will block warm-adapted species' colonization and cold-adapted species' emigration under warming. The three distinct patches signify a fragmented background and the community in the middle of the three patches was selected to exhibit colonization-extinction dynamics in fragmented habitats. Relative species richness is shown by the number of bird silhouettes in the community (d). Note that extinction here may include both the emigration of species and then the local extinction of species and that CTI can also change simply due to differential colonization-extinction rates by thermal affinity if the system is not at equilibrium prior to the study. In our study system, we have no way of knowing whether our island system was at equilibrium at onset of the study, thus, focusing on changing rates of colonization-extinction over time presents a much stronger tests of thermophilization.

to climate change are substantially reduced (Suggitt et al., 2018). We thus expect that the reduction in habitat heterogeneity associated with higher fragmentation will intensify the loss of cold-adapted species and weaken the colonization of warm-adapted species (**Figure 1c**).

Third, habitat fragmentation could prevent or interrupt species' dispersal which is essential for them to track suitable climatic niches (McGuire et al., 2016; Opdam & Wascher, 2004; Thomas et al., 2006). This limitation may come from higher mortality rate with increasing dispersal distance during searching for suitable patches (Brooker et al., 1999), and will influence the colonization-extinction dynamics underlying climate response (Fourcade et al., 2021; Hylander & Ehrlén, 2013). For example, habitat fragmentation renders habitats to be too isolated to be colonized, causing sedentary butterflies to lag more behind climate warming in Britain than mobile ones (Warren et al., 2001). All things being equal, less isolated patches should experience increasing rate in colonization of warm-adapted species under warming, benefitting from near sources of species; meanwhile, cold-adapted species should experience higher extinction rates on less isolated patches because they can emigrate easily from these patches under climate warming (**Figure 1d**).

Although the overall effect of habitat fragmentation on climate-induced community dynamics has been established, few empirical studies have ever considered these multiple mechanisms simultaneously. Here, we use 10 years of bird community data in a subtropical land-bridge island system (Thousand Island Lake, TIL, China, **Figure 2**) during a period of consistent climatic warming (**Appendix 1—figure 1**) to explore whether and how community thermal niche composition responds to climate change and how habitat fragmentation mediates the process. This reservoir island system was formed 65 years ago due to dam construction. Specifically, we focused on the following three predictions:

1. We predict the existence of thermophilization of bird communities in our fragmented land-bridge island system, characterized by an increasing temporal trend of CTI.
2. We predict that thermophilization of bird communities should result from both increasing warm-adapted species and abundances and decreasing cold-adapted species and abundances. We also predict that, over time with a warming climate, warm-adapted species should increase their colonization rate while cold-adapted species should increase their extinction rate (**Figure 1a**).
3. We predict that habitat fragmentation – measured as island area and distance to the mainland – will mediate the two main driving processes of thermophilization: colonization of warm-adapted species and extinction of cold-adapted species. Following our theoretical framework, if the climate buffering effect dominates (**Figure 1b**) area effects, we expect the differential trends of warm-adapted species versus cold-adapted species to be more extreme on smaller islands (i.e., even faster colonization of warm-adapted, even faster extinction of cold-adapted) due to weaker microclimate buffering. Alternatively, if the effects of habitat heterogeneity dominate (**Figure 1c**) area effects, cold-adapted species will again show greater extinction, but warm-adapted species will alternatively experience lower colonization on smaller islands (relative to larger islands) due to less abundant resources (Si et al., 2017). Finally, we predict that on more isolated islands (**Figure 1d**), the colonization of warm-adapted species and the extinction of cold-adapted species will increase more slowly because isolation hinders both immigration and emigration.

2 Results

The number of species detected in surveys on each island across the study period averaged 13.37 ± 6.26 (mean $\pm$ SD) species, ranging from 2 to 40 species, with an observed gamma diversity of 60 species. The STI of all 60 birds averaged 19.94 ± 3.58 °C (mean $\pm$ SD) and ranged from 9.30 °C

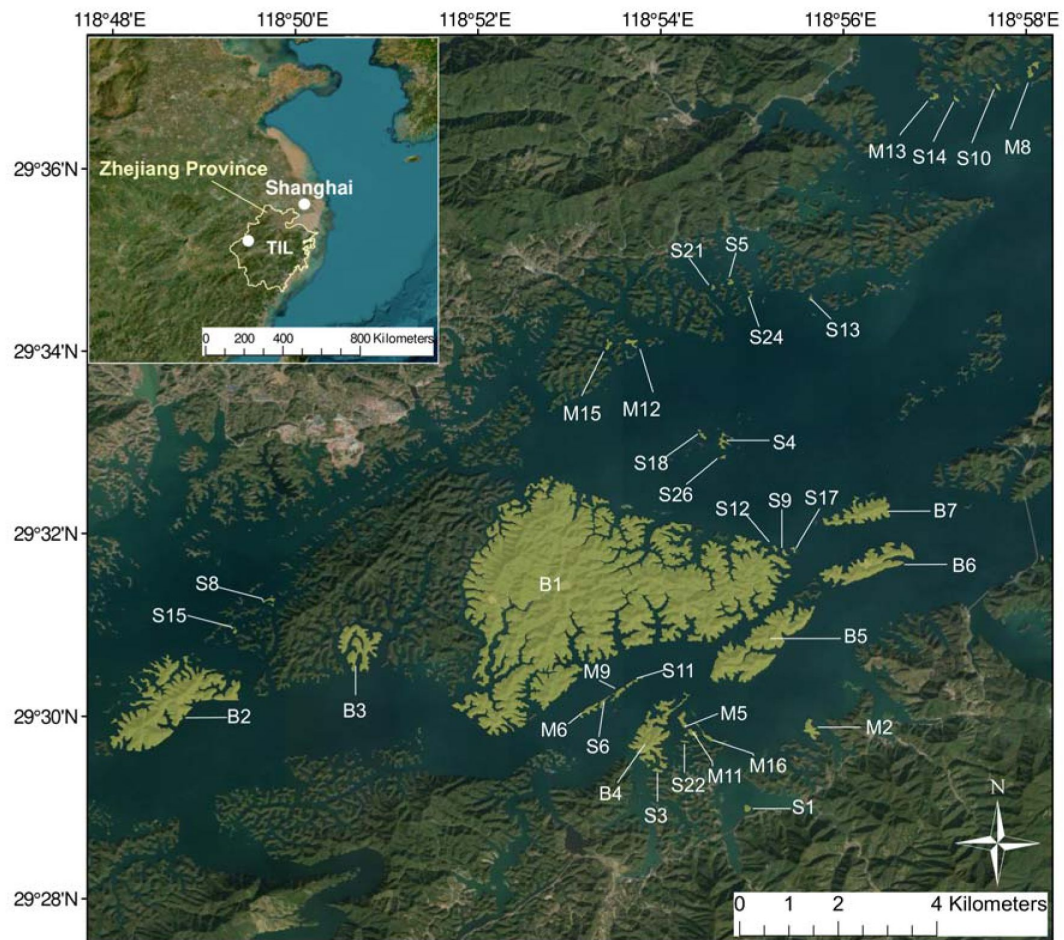


Figure 2.

The location of Thousand Island Lake and the experimental islands.

The map was created using ESRI (Environmental Systems Resource Institute) ArcMap software (version 10.3). The base map sources include Esri, Maxar, Earthstar Geographics, and the GIS User Community.

(*Cuculus canorus*) to 27.20 °C (*Prinia inornate*), with a median of 20.63 °C (**Appendix 1—figure 2** [↗](#); **Appendix 1—figure 3** [↗](#)). STI of resident species ($n = 47$) and summer visitors ($n = 13$) did not show a significant difference (t-test: $t = 0.23$, $df = 17.82$, $p = 0.82$). No significant correlation was found between STI and species' ecological traits; specifically, the continuous variables of dispersal ability, body size, body mass and clutch size (Pearson correlations for each, $|r| < 0.22$), and the categorical variables of diet (carnivorous/omnivorous/herbivory), active layer (canopy/mid/low), and residence type (resident species/summer visitor).

2.1 Thermophilization of bird communities

At the landscape scale, considering species detected across the study area, occurrence-based CTI (CTI_{occ}; see **section 4.4** [↗](#)) showed no trend (posterior mean temporal trend = 0.414; 95% CrI: -12.751, 13.554) but abundance-based CTI (CTI_{abun}; see **section 4.4** [↗](#)) showed a significant increasing trend (temporal trend [mean $\pm$ SE] = 0.327 ± 0.041 , $t = 7.989$, $p < 0.0001$). When measuring CTI trends for individual islands (expressed as %/unit year), we found significant increases in CTI for both occurrence- (mean temporal trend = 0.124; 95% CrI: 0.108, 0.137) and abundance-based indices (temporal trend = 0.320 ± 0.021 , $t_{493} = 15.534$, $p < 0.0001$; **Figure 3** [↗](#)).

2.2 Mechanisms underlying CTI trends

Comparing across species with different thermal affinities, we found a weak positive linear relationship between STI and species-specific temporal occupancy trends (t-test: slope = 0.206, $t_{58} = 1.766$, $p = 0.082$; **Figure 4a** [↗](#)), indicating warm-adapted species were marginally more likely to increase in occurrence over time. Decomposing occurrence change to dynamic parameters; however, we found a significant positive relationship between STI and species' temporal trends in colonization (t-test: slope = 0.259, $p = 0.043$) and a significant negative relationship between STI and species' temporal trends in extinction (t-test: slope = -0.414, $p = 0.015$; **Appendix 1—figure 4** [↗](#)). Thus, warm-adapted species generally increased in occupancy over time, which was driven by an increase in colonization probability and a decrease in extinction probability; in contrast, cold-adapted species generally decreased in occupancy over time, which was driven by an increase in extinction probability and a decrease in colonization probability (**Figure 4b** [↗](#)). Considering all species, the influence of a thermal-association gradient underlying occurrence dynamics resulted in a particularly strong correlation between species' trends in colonization and extinction probabilities (Pearson correlation $r = -0.77$; **Figure 4b** [↗](#)).

Similar patterns were found for abundance, where we detected a strong positive relationship (t-test: slope = 0.153, $t_{58} = 3.047$, $p = 0.003$) between STI and each species' temporal trend in abundance: warm-adapted species generally increased abundance while cold-adapted species decreased over 10 years (**Figure 4c** [↗](#)). At the island scale, both total abundance and total species richness increased significantly for the warm-adapted species group while both responses significantly decreased for the cold-adapted species (richness trend, $p < 0.0001$; total abundance trend, $p < 0.0001$; **Appendix 1—figure 5** [↗](#)).

2.3 The mediating effects of fragmentation on colonization and extinction processes

Both colonization and extinction probabilities showed evidence of being moderated by fragmentation (**Figure 5** [↗](#)). Of particular interest was how the temporal trend in dynamic parameters varied as a function of area or isolation (i.e., Year:Area or Year:Isolation effects). Although species-specific interaction terms were not generally significant for most species, overall trends could be assessed across species. In particular, the effect of isolation on temporal dynamics of thermophilization was relatively consistent across cold- (**Figure 5a** [↗](#)) and warm-adapted species (**Figure 5b** [↗](#)); specifically, on islands nearer to the mainland, warm-adapted species (15

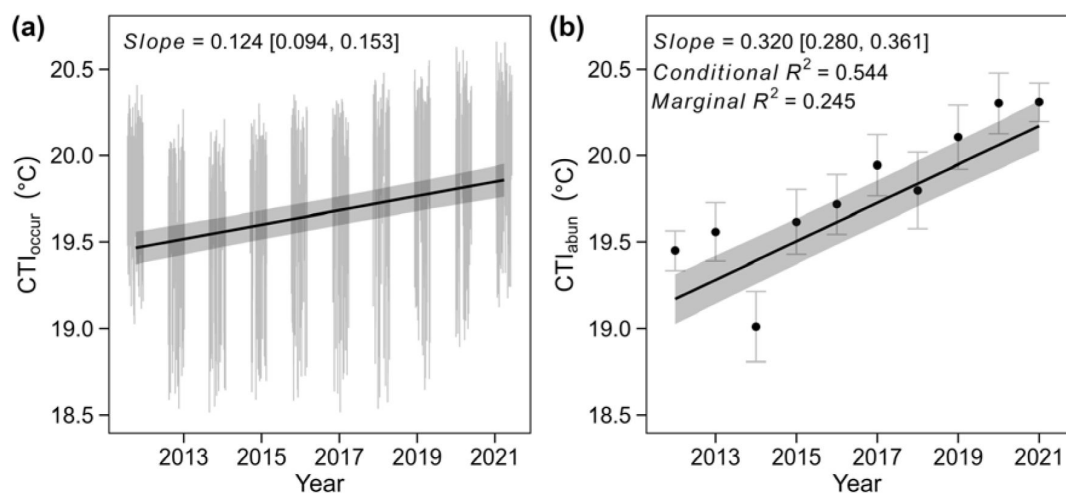


Figure 3.

Temporal trend in CTI_{occ} and CTI_{abun} in the TIL.

Vertical lines in **(a)** are posterior estimated CTI (mean $\pm$ SD); solid black line and shaded area are predictions and 95% credible intervals extracted from JAGS modeling posterior mean CTI_{occ} as a function of year, and using island identity as a random effect while accounting for the variation in posterior CTI_{occ} . Points and error bars in **(b)** are observed CTI_{abun} (mean $\pm$ 2SD); solid black line and shaded area are predicted values and 95% confidence intervals estimated from LMM modeling observed CTI_{abun} as a function of year, and using island identity as a random effect.

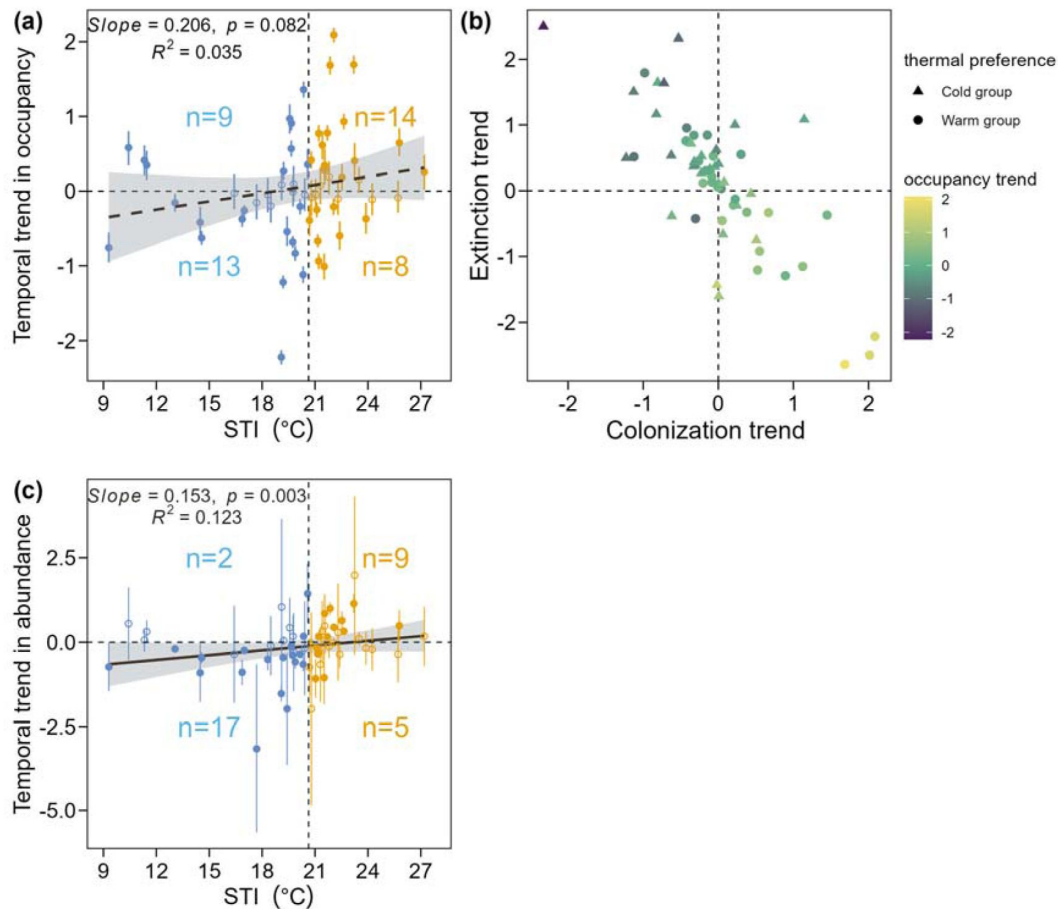


Figure 4.

Relationship between STI and occupancy trend, abundance trend and the relationship among species thermal preference and trend in occupancy, colonization, and extinction rates.

Each point and error bar in **(a)** represents the temporal trend of occupancy rate (posterior mean and 95% credible interval of year effect on occupancy rate) for each species, with the filled dots indicating a significant year effect while the hollow dots indicating nonsignificant year effect. The dotted vertical line indicates the median of STI values. Cold-adapted species are plotted in blue and warm-adapted species are plotted in orange. The number of species with significant occupancy trends in each quadrant was added to the plot. The black dashed line and shaded area are the predicted values and 95% confidence intervals of the weighted linear regression model. Each point in **(b)** represents the posterior mean estimate of year in colonization, extinction or occupancy rate for each species. The color of the point indicates the temporal trend in occupancy. Similar to **(a)**, each point and error bar in **(c)** represents the temporal trend of abundance (year effect and 95% confidence interval of year effect on abundance) for each species.

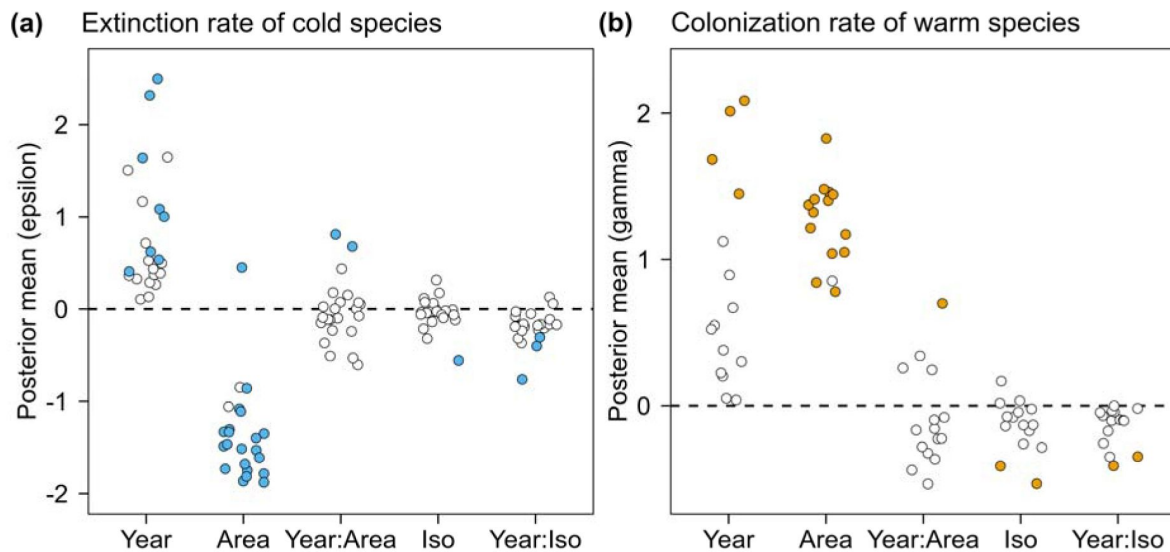


Figure 5.

Posterior estimates of logit-scale parameters related to cold-adapted species' extinction rates and warm-adapted species' colonization rates.

Points are species-specific posterior means on the logit-scale, where parameters > 0 indicate positive effects (on extinction [a] or colonization [b]) and parameters < 0 indicate negative effects. Only cold-adapted species with positive trends in extinction (a) and warm-adapted species with positive trends in colonization (b) are shown, which are two main processes contributing positively to thermophilization. Points in blue/orange indicate significant effects (95% or 80% credible intervals) and points in white indicate non-significant effects. All points were jittered slightly by 0.22 in width.

out of 15 investigated species) increased their colonization probability at a higher rate over time, while most cold-adapted species (21 out of 23 species) increased their extinction probability at a higher rate.

Island area, in contrast, had a more mixed role in moderating colonization/extinction rates across species. For most warm-adapted species (11 out of 15 investigated species; [Figure 5b](#)), colonization rates increased faster over time on smaller islands (i.e., negative area:year interaction term); however, the colonization rates of the other four species increased faster on larger islands, including the only significant positive interactive effect (*Dendrocitta formosae*). For cold-adapted species, extinction rates were approximately equally split between increasing faster on smaller islands (12 out of 23 species) and increasing faster on larger islands (11 out of 23 species), including two species, *Spizixos semitorques* and *Pericrocotus cantonensis*, which showed significantly increasing rates of extinction on larger islands.

3 Discussion

Our study explored the driving processes underlying a decade of thermophilization of a bird community breeding in a model fragmented system, specifically testing whether and how habitat fragmentation may mediate the colonization-extinction dynamics which underly shifts in mean community traits. Our results confirm that community thermal niche composition was changing directionally, mainly through gains in warm-adapted species and individuals and losses in cold-adapted species and individuals. We further found that habitat fragmentation influences two processes of thermophilization: colonization rates of most warm-adapted species tended to increase faster on smaller and less isolated islands, while the loss rates of most cold-adapted species tended to be exacerbated on less isolated islands.

3.1 Thermophilization of bird communities

Although thermophilization has been found in many taxa ([Lajeunesse & Fourcade, 2022](#)) and is more apparent in the tropics than in temperate zones ([Freeman et al., 2021](#)), there are few empirical demonstrations of this process in subtropical fragmented systems. We found decadal-scale thermophilization evident in both occurrence- and abundance-based metrics, although the response was considerably stronger for abundance. This result is consistent with work in other systems, where abundance-weighted CTI generally responds more quickly than occurrence-based CTI given climatic warming ([Devictor et al., 2008](#); [Lindström et al., 2013](#); [Oliver et al., 2017](#); [Tayleur et al., 2016](#)). Notably, when tested on the landscape scale (versus on individual island communities), only the abundance-based thermophilization trend was significant, indicating thermophilization of bird communities was mostly due to occurrence dynamics within the region, rather than exogenous community turnover outside the region.

3.2 Mechanisms underlying CTI trends

Consistent with our expectation, we found an overall increase in occupancy and abundance of warm-adapted species and a decrease in occupancy and abundance of cold-adapted species, which together contributed to thermophilization. This pattern is consistent with many previous studies conducted in continuous habitats in Europe or North America both for breeding and non-breeding bird communities ([Curley et al., 2022](#); [Prince & Zuckerberg, 2015](#); [Tayleur et al., 2016](#)). Also consistent with our expectation was that occupancy change was the product of temporal trends in both colonization and extinction dynamics; namely, warm-adapted species showed increasing colonization and decreasing extinction rates, while, cold-adapted species showed decreasing colonization and increasing extinction rates. Following that STI is predictably related to temporal trends in colonization and extinction rate ([Figure 4](#)), we can conclude that STI is an effective broad indicator of relative climatic vulnerability within a community, where species with

relatively higher thermal preferences are generally poised to benefit from local climate warming and vice versa. This conclusion is supported by studies that have established relationships between STI and species' population trend (Pearce-Higgins et al., 2015 [↗](#); Rigal et al., 2023 [↗](#)), species' contribution analysis (Tayleur et al., 2016 [↗](#)), or spatial occupancy (Anderson et al., 2023 [↗](#)). Decomposing average community trait shifts into colonization and extinction dynamics of individual species and their associated traits provides a comprehensive and more nuanced insight into community response to climate change.

There are numerous reasons why we would expect temperature to strongly drive community change, as temperature is related to species' survival through food resources, foraging time, and thermoregulatory requirements (Dunn & Møller, 2019 [↗](#)). The local population growth of warm-adapted species may mirror their range expansion under climate warming. For example, thermophilization of avian communities in North America is mainly owed to the species that have increased in both range and population size (Curley et al., 2022 [↗](#)). Contemporary latitudinal range shifts are thought to be primarily resultant of the expansion of polar margins of equatorial-distributed species (La Sorte & Thompson, 2007 [↗](#); Thomas & Lennon, 1999 [↗](#)), possibly due to extinction lags at equatorial margins (La Sorte & Jetz, 2012 [↗](#)). TIL is near the cold edge of most warm-adapted species like *Hemixos castanonotus*, *Pericrocotus solaris*, and *Lonchura punctulate*. Combined with the prediction that most birds in China should move upslope or poleward under climate change (Hu et al., 2020 [↗](#)), we infer that local climate will continue to shift these warm-adapted species toward a thermal optimum, predicting continued population growth in the near future for these species (Sagarin & Gaines, 2002 [↗](#)).

Meanwhile, most cold-adapted species in our system demonstrated increasing extinction rates over time. Extinctions are more difficult to detect than range expansions, partly due to temporal lags (Thomas et al., 2006 [↗](#); Thomas & Lennon, 1999 [↗](#)) but also due to pervasive bias of false absences in time series data (Tingley & Beissinger, 2009 [↗](#)). Moreover, species' extinctions are more likely to be mediated by factors such as rainfall, species interactions, and changes in vegetation (Paquette & Hargreaves, 2021 [↗](#); Thomas & Lennon, 1999 [↗](#)), which will further lead to unequal rates of extinction relative to colonization. In our community, although there are many cold-adapted species – such as *Streptopelia orientalis*, *Phasianus colchicus*, *Passer montanus*, and *Emberiza cioides* – that have northern distributional limits much farther northern than our site, our site is not close to the current southern boundary for these species either. Even so, they still showed considerable loss in TIL, suggesting fragmented landscape exacerbated the negative effect of warming on these cold-adapted species.

Despite the general trend, not all warm-adapted species increased (in abundance or occupancy) and not all cold-adapted species decreased. In other words, not all species contributed equally to the process of thermophilization, as has been seen elsewhere (Tayleur et al., 2016 [↗](#)). Variation in species' response relates to their true thermal tolerance in the study region, which would require additional physiological data to measure and could also be influenced by habitat usage and biotic interactions.

3.3 The mediating effect of fragmentation on colonization and extinction processes

It has long been thought that habitat fragmentation can impact species' climate tracking (Opdam & Wascher, 2004 [↗](#)) but there are only a few empirical studies to date (Fourcade et al., 2021 [↗](#); Warren et al., 2001 [↗](#)). Our research examining colonization-extinction dynamics provides empirical evidence of the possible mechanisms in a land-bridge island system.

As expected, the mediating effect of isolation on the two main processes of thermophilization was relatively consistent, with warm-adapted species colonizing faster and cold-adapted species being extirpated faster on islands nearer to the mainland. This indicates that isolation from population

sources can limit the dispersal that underlies range shifts, supporting the general perception that habitat fragmentation is thought to impede climate-driven range expansions (Opdam & Wascher, 2004) or even block them (Warren et al., 2001). While we cannot truly distinguish in our system between local extinction and emigration, we suspect that given two islands equal except in isolation, if both lose suitability due to climate change, individuals can easily emigrate from the island nearer to the mainland, while individuals on the more isolated island would be more likely to be trapped in place until the species went locally extinct due to a lack of rescue.

As a caveat, we only consider the distance to the nearest mainland as a measure of fragmentation, consistent with previous work in this system (Si et al., 2014), but we acknowledge that other distance-based metrics of isolation that incorporate inter-island connections could reveal additional insights on fragmentation effects. The spatial arrangement of islands, like the arrangement of habitat, can influence niche tracking of species (Fourcade et al., 2021). Future studies should take these metrics into account to thoroughly understand the influence of isolation and spatial arrangement of patches in mediating the effect of climate warming on species.

Island size also had pervasive effects on underlying thermophilization dynamics. In our study, the colonization rate of a large proportion of warm-adapted species (11 out of 15) and half of cold-adapted species (12 out of 23) were increasing more rapidly on smaller islands. Similarly, a lower proportion of habitats enhances the colonization of warm-adapted species under climate change in the butterfly assemblage (Fourcade et al., 2021). The observed colonization-extinction patterns on smaller islands support the hypothesis that warm-adapted species benefit from the lowered thermal buffering of ambient temperature on smaller islands, while the same mechanism may speed up extinction (or emigration) of cold-adapted species (Figure 1b). To investigate this mechanistic hypothesis more thoroughly, using understory air temperature data monitored on 20 islands in the breeding season in 2022, we found that temperature buffering ability indeed was diminished on smaller islands (Appendix 1—figure 6), further supporting microclimate as a primary driver of fragmentation's mediation of thermophilization (Zellweger et al., 2020).

The increased extinction rate of some cold-adapted species on smaller islands is also consistent with the hypothesized impacts of reduced habitat heterogeneity (Figure 1c). Previous studies found that areas with greater amounts of habitat featured lower extinction rates of cold-adapted butterfly species (Fourcade et al., 2021), possibly by facilitating movement to colder microclimates (Suggitt et al., 2018). We thus suppose that habitat heterogeneity could also mitigate the loss of these relatively cold-adapted species as expected. Habitat diversity, including the observed number of species, the rarefied species richness per island, species density and the rarefied species richness per unit area, all increased significantly with island area instead of isolation in our system (Liu et al., 2020). Thus, as habitat heterogeneity was only consistent with our observed response of cold-adapted species (Figure 1c), while microclimate buffering is consistent with observed responses of both warm- and cold-adapted species (Figure 1b), our evidence more consistently supports the latter than the former.

Contrary to any expected mechanisms (Figure 1), the extinction rate of some cold-adapted species – such as *Spizixos semitorques* and *Pericrocotus cantonensis* – was increased faster on larger islands. Although post hoc, we speculate that these species face stronger predation pressures on larger islands. For example, predators like snakes and felids may also increase activity under climate warming (DeGregorio et al., 2015), but these predators only exist in our system on the larger islands. Besides, *Garrulax canorus*, which is a warm-adapted species, increased faster on larger islands. This species may depend more on habitat heterogeneity than microclimate alone, including food resources and microrefugia, which is related to faster climate tracking (Gäüzère et al., 2017). Overall, these idiosyncratic responses reveal several possible mechanisms in regulating species' climate responses, including resource demands and biological

interactions like competition and predation. Future studies are needed to take these factors into account to understand the complex mechanisms by which habitat loss mediates species range shifts.

Overall, our findings have important implications for conservation practices. Firstly, we confirmed the role of isolation in limiting range shifting. Better connected landscapes should be developed to remove dispersal constraints and facilitate species' relocation to the best suitable microclimate. Second, small patches can foster the establishment of newly adapted warm-adapted species while large patches can act as refugia for cold-adapted species. Therefore, preserving patches of diverse sizes can act as stepping stones or shelters in a warming climate depending on the thermal affinity of species. These insights are important supplement to the previous emphasis on the role of habitat diversity in fostering (Richard et al., 2021 [↗](#)) or reducing (Gaüzère et al., 2017 [↗](#)) community-level climate debt.

4 Materials and Methods

4.1 Study area and islands selection

The Thousand Island Lake (TIL), located in eastern China, was formed in 1959 following the construction of the Xin'anjiang Dam for hydroelectricity (Figure 2 [↗](#)). When the lake is at its highest, there are 1078 islands with an area larger than 0.25 ha. Currently, about 90% of the forested areas are dominated by Masson pine (*Pinus massoniana*) in the canopy and broad-leaved species (e.g. *Loropetalum chinense*, *Vaccinium carlesii* and *Rhododendron simsii*) in the sub-canopy and understory (Liu et al., 2020 [↗](#)). The climatic zone is a typical monsoon climate. The precipitation is mainly concentrated between April and June with an average yearly rainfall of 1430 mm. The average annual temperature is 17°C (hottest from June to August) and the average daily temperature ranges from −7.6 °C in January to 41.8 °C in July (Si et al., 2017 [↗](#)).

We selected 36 islands according to a gradient of island area and isolation with a guarantee of no significant correlation between island area and isolation (Pearson $r = -0.21$, $p = 0.21$). For each island, we calculated island area and isolation (measured in the nearest Euclidean distance to the mainland) to represent the degree of habitat fragmentation (Figure 2 [↗](#)). Distance to the mainland is the best distance-based measure in fitting species' colonization rate and extinction rate in TIL (Si et al., 2014 [↗](#)). Since lake formation, the islands have been protected by forbidding logging, allowing natural succession pathways to occur.

4.2 Bird data

We established 53 survey transects on 36 islands with the sampling effort on each island roughly proportional to the logarithm of the island area (Schoereder et al., 2004 [↗](#)). As a result, 53 transects on 36 islands were sampled, including eight transect trails on the largest study island (area = 1058 ha), four transects on two islands between 100 and 500 ha, two transects on four islands between 10 and 100 ha, and one on each of the remaining small islands (c. 1 ha for most islands) (Si et al., 2018 [↗](#)). Breeding bird communities were surveyed on each transect nine times annually (three times per month from April to June) from 2012 to 2021 (Si et al., 2017 [↗](#)). In each survey, observers walked along each transect at a constant speed (2.0 km/h) and recorded all the birds seen or heard on the survey islands. To minimize the bias, the order and direction of each island surveyed were randomized. We based our abundance estimate on the maximum number of individuals recorded across the nine annual surveys. We excluded non-breeding species, nocturnal and crepuscular species, high-flying species passing over the islands (e.g., raptors, swallows) and strongly water-associated birds (e.g., cormorants) from our record. First, our surveys were conducted during the day, so some nocturnal and crepuscular species, such as the owls and nightjars were excluded for inadequate survey design. Second, wagtail, kingfisher, and water birds such as ducks and herons were excluded because we were only interested in forest

birds. Third, birds like swallows, and eagles who were usually flying or soaring in the air rather than staying on islands, were also excluded as it was difficult to determine their definite belonging islands. Following these operations, 60 species were finally retained.

4.3 Climate data

We obtained climate data (monthly average temperature from 2012 to 2021) from the Meteorological Bureau of Chun'an County in Zhejiang Province, China. We calculated breeding season temperature as the average of the mean monthly temperature from April to June in order to represent the thermal conditions birds experience at this site each year (Lindström et al., 2013). Breeding season temperature increased significantly over 10 years (slope = 0.078, $r^2 = 34.53\%$, $p = 0.04$, **Appendix 1—figure 1**).

4.4 STI and CTI

We followed the methods of Devictor et al. (2008) to calculate a species temperature index (STI) using ArcMap 10.3. STI of a given species is defined as the average temperature of breeding months (April – June) across its distributional range (restricted to the Northern hemisphere) averaged over 1970–2000. Monthly temperature data were obtained from WorldClim at a resolution of 30-arc seconds (<http://www.worldclim.org>). Distributional maps were extracted from Birdlife International 2019 (<http://datazone.birdlife.org/species/requestdis>), in which we selected only the distributional regions where the species is either resident or breeding. All species were divided into two groups indicating their relative thermal preference: species with STI higher than the median STI were labeled as warm-adapted species, and species lower than the median STI were labeled as cold-adapted species (A. E. Bates et al., 2017). The STI of a species can differ depending on distributional and climate data but the rank order of species STI is highly correlated across methods (Barnagaud et al., 2013). We verified the robustness of our STI calculations using different distributional ranges and annual time windows (**Appendix 1—figure 2**).

CTI is a community-level index representing the average species thermal niche of all species or individuals in the community (Devictor et al., 2008). Accordingly, for each community in each year, CTI was calculated as the average STI of all occurring species (hereafter: CTI_{occur}) or counted individuals (hereafter: CTI_{abun}):

$$CTI_{\text{occur},j,t} = \frac{\sum_{i=1}^{N_{j,t}} STI_i}{N_{j,t}}$$

$$CTI_{\text{abun},j,t} = \frac{\sum_{i=1}^{N_{j,t}} (STI_i \times A_{i,j,t})}{\sum_{i=1}^{N_{j,t}} A_{i,j,t}}$$

where $N_{j,t}$ is the total number of species surveyed in the community j in year t , $A_{i,j,t}$ denoted the maximum abundance among nine surveys of the i^{th} species in community j in year t .

CTI_{occur} was calculated using occurrence data corrected for imperfect detection (see next section), while CTI_{abun} was estimated using the maximum annual count across 9 surveys. Given that our survey effort was so high (i.e., 9 repeat surveys per year), true abundance should be highly correlated with maximum observed abundance (MacKenzie et al., 2006). We also used these occurrence and abundance metrics from all islands to compute each year's regional CTI at the landscape level.

4.5 Multispecies dynamic occupancy model

Presence-absence data can be highly sensitive to false negatives, so to explore the driving processes of change in CTI_{occu} , and to explore how habitat fragmentation mediated the main processes while accounting for the imperfect detection, we developed a spatially hierarchical dynamic multi-species occupancy model (MSOM) in a Bayesian framework based on the model by Royle and Kéry (2007) [\[1\]](#). We denote y_{nijt} as the observation (detected = 1; undetected = 0) for species n (1–60 species) in survey visit j (1–9 surveys) at transect i (1–53 transects) in year t (1–10 years). Observation, y_{nijt} is assumed to be the result of imperfect detection of the true occurrence status, z_{nit} (1 or 0), and is thus modeled as a Bernoulli-distributed variable with a probability of $p_{nijt} \times z_{nit}$, where p_{nijt} is the probability of detection for a given survey along a transect:

$$y_{nijt} \sim \text{Bernoulli}(z_{nit} * p_{nijt})$$

where z_{nit} is assumed to be of fixed presence/absence status for a given species across all j survey intervals within year t . However, given that individual birds move dynamically across territories within a year, during 9 surveys each year, we broadly interpret our estimates of “occupancy” as “use” (Si et al., 2018 [\[2\]](#)) to relax the closure assumption (MacKenzie et al., 2004 [\[3\]](#)).

Each site has an initial value of occupancy (ψ_{ni1}) which can be given a uniform prior distribution from 0 to 1, and occupancy (ψ_{nit}) in subsequent years ($t > 1$) is determined based on whether sites become colonized or remain occupied through persistence (MacKenzie et al., 2003 [\[3\]](#)):

$$Z_{nit} \sim \text{Bernoulli}(\psi_{nit})$$

$$\psi_{nit} = Z_{ni,t-1}(1 - \epsilon_{ni,t-1}) + (1 - Z_{ni,t-1})\gamma_{ni,t-1} \text{ for } t > 1$$

where ψ_{nit} , $\epsilon_{ni,t-1}$, $\gamma_{ni,t-1}$ are transect-level probabilities of occupancy, extinction, and colonization, respectively.

We modeled the probability of detection, p_{nijt} , as a function of the length of the transect, $length_i$ and the ordinal day of year, day_{ijt} . Island identity was included as a random effect to account for the nonindependence of transects within the same islands.

$$\text{logit}(p_{nijt}) = p0_n + p1_n \text{length}_i + p2_n \text{day}_{ijt} + \text{island}_{REni}$$

To explore how colonization rate and extinction rate change across 10 years and how habitat fragmentation mediated these dynamics processes, we modeled site-level extinction and colonization ($\epsilon_{ni,t}$, $\gamma_{ni,t}$) each as a logit-linear function of 5 covariates: year, island area, isolation, the interaction between year and area and between isolation and year. Year was added as a random slope, thus allowing the temporal trend in the colonization or extinction to vary with island area and isolation:

$$\text{logit}(\epsilon_{ni,t}) = \alpha0_n + \alpha1_n \text{area}_i + \alpha2_n \text{isolation}_i + (\alpha3_n + RE_i) \text{year}_t + \alpha4_n \text{area}_i \text{year}_t + \alpha5_n \text{isolation}_i \text{year}_t + \text{island}_{REni}$$

$$\text{logit}(\gamma_{ni,t}) = \beta0_n + \beta1_n \text{area}_i + \beta2_n \text{isolation}_i + (\beta3_n + RE_i) \text{year}_t + \beta4_n \text{area}_i \text{year}_t + \beta5_n \text{isolation}_i \text{year}_t + \text{island}_{REni}$$

In all cases, continuous covariates were z-transformed (mean of 0 and a standard deviation of 1). There is no strong correlation between island area and isolation (Pearson correlation $r = -0.214$) so these effects can be considered independent. We fit the MSOM with JAGS (Plummer, 2003 [\[4\]](#)) using

the package *rjags* (Plummer, 2023 [↗](#)) in R v4.3.1 (R Core Team, 2023 [↗](#)). We used vague priors (e.g., normal with $\mu = 0$, $\tau = 0.01$). We ran three chains for 60,000 iterations, discarded the first 40,000 as burn-in and thinned by 20, yielding a combined posterior sample of 3000. We extracted 600 iterations across three chains for Z to calculate CTI_{occur} . Convergence was checked visually with trace plots and confirmed with a Gelman–Rubin statistic < 1.1 (Gelman & Rubin, 1992 [↗](#)). Inference on parameters was made using 95% and 80% Bayesian credible intervals. We checked the posterior predictive ability of the model fit by calculating Bayesian p -values ($p = 0.488$, indicating an unbiased estimation of our model) (Gelman et al., 1996 [↗](#)).

4.6 Statistical analysis

4.6.1 Thermophilization of bird communities

To test for occurrence-based thermophilization at the island level, Bayesian linear regression using *R2jags* (Su & Yajima, 2021 [↗](#)) was used to derive a posterior estimate of the temporal trend in CTI_{occur} while propagating error in the estimation of CTI_{occur} . The continuous variable *year* (z-transformed) was incorporated as the only predictor. To account for the nonindependence of data within islands, we included island identity as a random intercept, thus allowing intercepts to vary across islands. We used vague priors (e.g., normal with $\mu = 0$, $\tau = 0.001$) for intercept and year effect. We ran three chains for 10,000 iterations, discarded the first 5,000 as burn-in and thinned by 5, yielding a combined posterior sample of 3000. We used the function *MCMCsummary* to get posterior intervals of the intercept and year effect, which were then used for plotting.

To test for abundance-based thermophilization, linear mixed effect models (LMM) with restricted maximum likelihood (REML) were used to model temporal trends in CTI_{abun} using function *lme* in R package *nlme* (Pinheiro et al., 2017 [↗](#)). As with the occurrence analysis, we used the continuous variable *year* (z-transformed) as the only fixed effect and island as a random intercept. To account for the temporal autocorrelation of CTI from the same island through time, we included the first-order correlation structure (e.g. the *corAR1* structure) because observations sampled from the nearby years may be more similar.

We also tested whether the overall regional bird composition (i.e. gamma diversity) experienced thermophilization. Bayesian linear regression was used for modeling temporal trends in regional CTI_{occur} . For abundance, a generalized least square (GLS) model was used to test the relationship between CTI_{abun} and continuous variable *year* while accounting for potential temporal autocorrelation. GLS was conducted using *gls* function in *nlme* R package (Pinheiro et al., 2023 [↗](#)).

4.6.2 Mechanisms underlying CTI trends

To explore the dynamics underlying trends in CTI_{occur} , specifically, whether thermophilization was accompanied by increasing occupancy of warm-adapted species and decreasing occupancy of cold-adapted species, we extracted 600 posterior samples of occupancy rate per species, and summarized ψ_{nt} by its mean and variation (logit-transformed to meet normality). We then modeled each species' temporal trend in occupancy (ψ_{nt}) over time using Bayesian linear regression in *R2jags* (Su & Yajima, 2021 [↗](#)) to account for propagated uncertainty in ψ_{nt} . Similarly, to decompose the CTI_{abun} trend, we modeled each species' temporal trend in maximum abundance using a generalized linear mixed effect model with Poisson error structure while accounting for potential overdispersion – via an observation-specific random effect (Harrison, 2014 [↗](#)). Weighted linear regression models tested for a relationship between STI and occupancy trend and between STI and abundance trend. This analysis was conducted using function *lmer* in package *lme4* (D. Bates et al., 2015 [↗](#)). Temporal correlations between occupancy and colonization or extinction rates were measured with Pearson correlations, using colonization and extinction rates derived directly from the fitted MSOM.

As a complementary analysis, we also investigated the trend in species richness or total abundance of the two thermal preference groups, separately. The total richness or abundance in each group was calculated on each transect and in each year. We then modeled temporal trends of natural logarithm-transformed richness or abundance using LMM with island identity as the random effect. Partial regression plots from LMM were produced using *ggeffects* R package (Lüdtke, 2018 [DOI](#)). For each model, we checked for normality of residuals and computed goodness-of-fit metrics (R^2) including conditional and marginal R^2 (Nakagawa & Schielzeth, 2013 [DOI](#)) using *performance* R package (Lüdtke et al., 2021 [DOI](#)). All continuous fixed effects in the models were standardized (mean = 0 and SD = 1) to ease computation and facilitate the interpretation and comparison of coefficients within and across models (Schielzeth, 2010 [DOI](#)).

4.6.3 The mediating effects of habitat fragmentation on colonization and extinction processes

An increasing colonization trend of warm-adapted species and increasing extinction trend of cold-adapted species are two main expected processes that cause thermophilization (Fourcade et al., 2021 [DOI](#)). To test our third prediction about the mediating effect of habitat fragmentation, we selected warm-adapted species that had an increasing trend in colonization rate (positive year effect in colonization rate) and cold-adapted species that had an increasing extinction rate (positive year effect in extinction rate). For each of these species, we extracted their posterior effect of year, area, isolation, and the interaction terms from previously fit models. The interaction terms reveal if habitat fragmentation positively or negatively affects either colonization or extinction processes.

Acknowledgements

We sincerely thank numerous graduate students in our group for the bird surveys in the field, and the Xin'an River Ecological Development Group Corporation, Chun'an Forestry Bureau, and the Thousand Island Lake National Forest Park for research permits. We also thank Dr. Di Zeng and Dr. Yuhao Zhao from East China Normal University for their insightful comments on the research. Zhe Liu from Ludong University helped a lot in learning ArcMap.

Additional information

Funding

This study is supported by the National Natural Science Foundation of China (#32030066 to P.D.; #32071545 to Xingfeng Si), Natural Science Foundation of Zhejiang Province (#LD21C030002 to Ping Ding). The funders had roles in study design, data collection and interpretation, and the decision to submit the work for publication.

Author contributions

Juan Liu: Conceptualization; data curation; formal analysis; methodology; visualization; writing – original draft; writing – review and editing. Morgan W. Tingley: formal analysis; methodology; visualization; writing – review and editing. Qiang Wu: data curation; visualization; writing – original draft (supporting); writing – review and editing. Peng Ren: data curation; visualization; writing – original draft (supporting); writing – review and editing. Tinghao Jin: data curation; formal analysis; methodology; visualization; writing – original draft (supporting); writing – review and editing. Ping Ding: Supervision; conceptualization; data curation; funding acquisition; methodology; writing – review and editing. Xingfeng Si: Supervision; conceptualization; methodology; data curation; funding acquisition; writing – review and editing.

Competing interests

The authors declare no conflicts of interests.

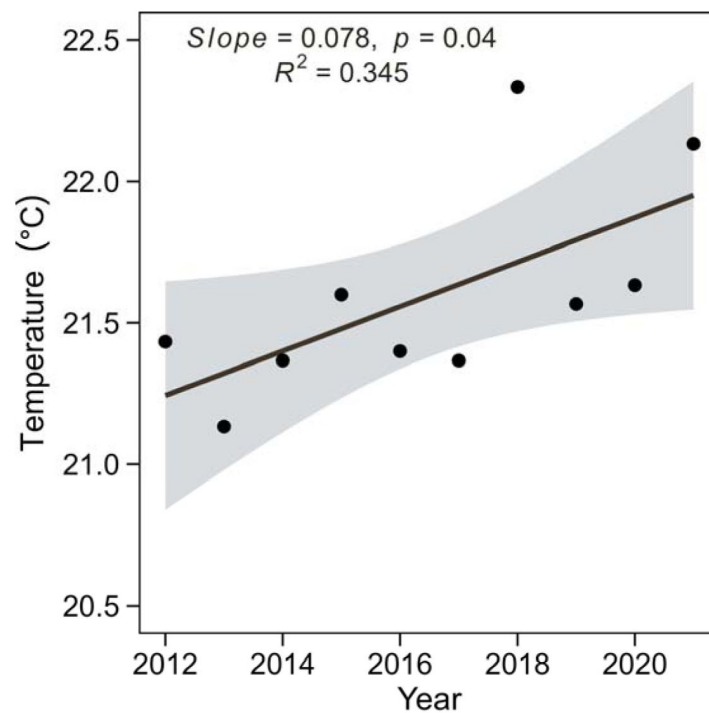
Data availability

The data that support the findings of this study are openly available in Figshare at <https://figshare.com/s/7a16974114262d280ef7> .

Code availability

The code that supports the findings of this study is openly available in Figshare at <https://figshare.com/s/7a16974114262d280ef7> .

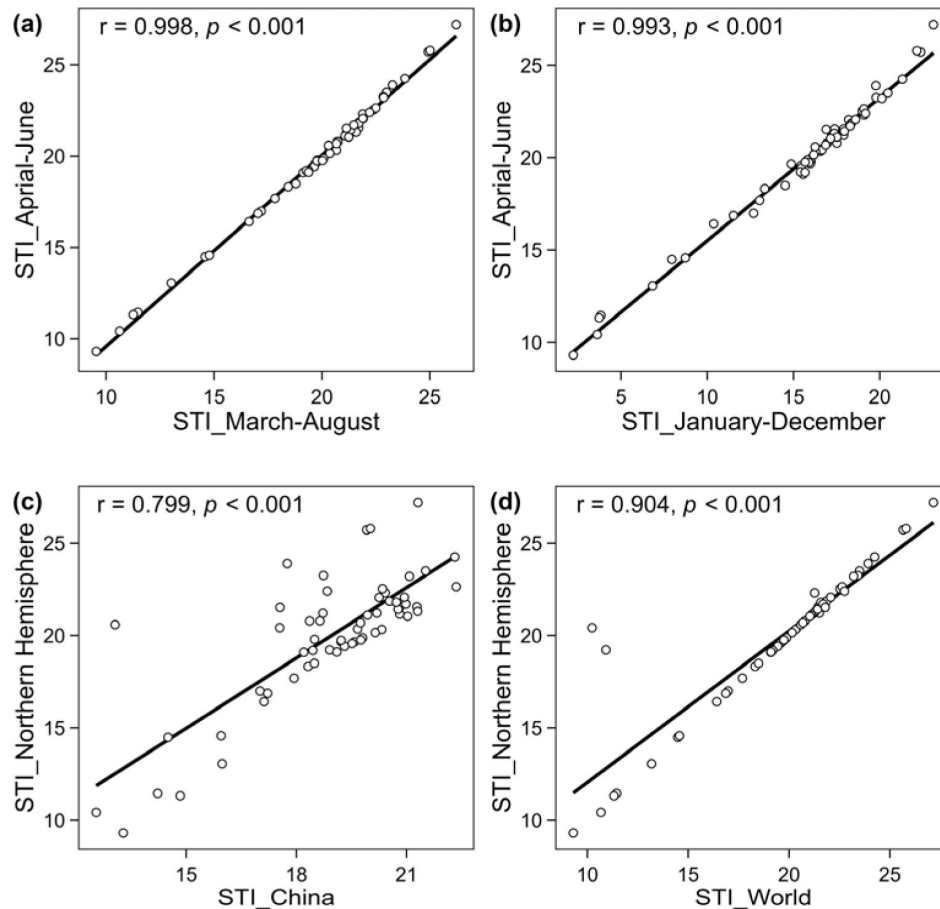
Appendix 1



Appendix 1—figure 1.

Breeding season temperature from the year 2012 to 2021 in the Thousand Island Lake, China.

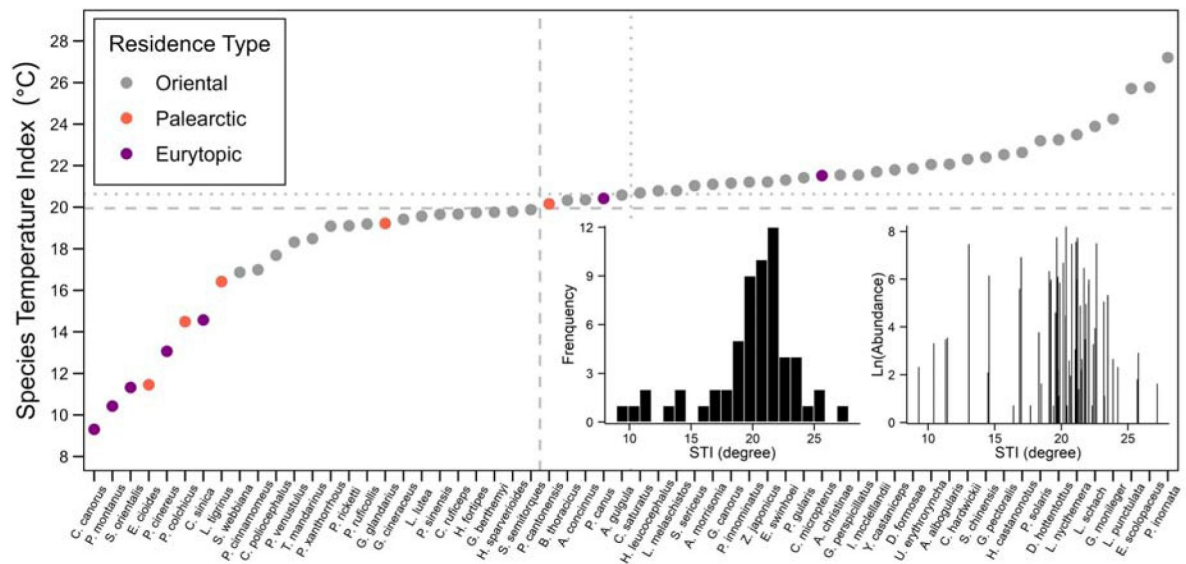
Breeding season temperature was calculated as the average of the mean monthly temperature for breeding seasons (April – June). Black lines and grey shade indicate the regression line and confidence interval of the linear regression model (slope = 0.078, $r^2 = 34.53\%$, $p = 0.04$).



Appendix 1—figure 2.

Robustness of STI to changes in data resources.

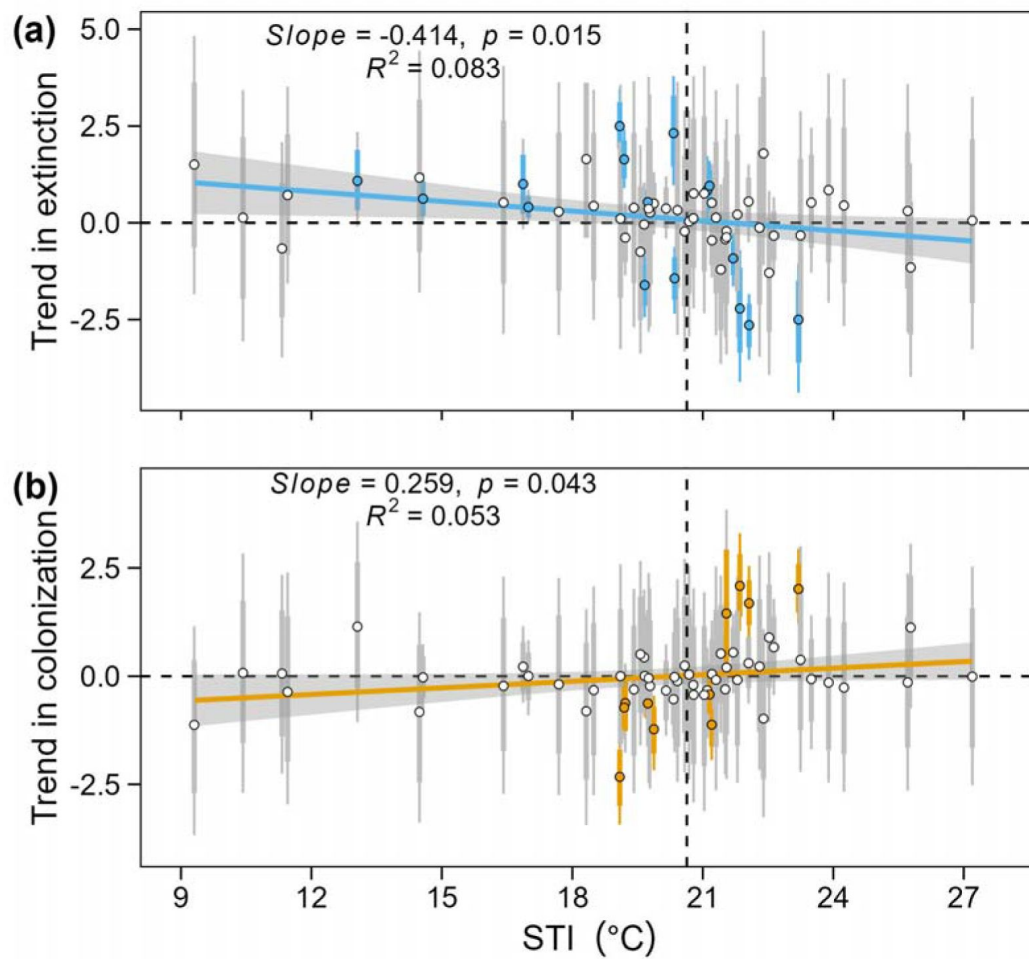
The units on the horizontal and vertical axes are degrees Celsius. Pearson correlation coefficients between STI used in our study and STI calculated with different time-windows: March to August (a) and January to December (b), were significantly high. Pearson correlation coefficients between STI used in our study and STI calculated with different distributional ranges: the distributional range in China (c) and distributional range across the world (d), were significantly high.



Appendix 1—figure 3.

Species Temperature Index of 60 species in the Thousand Islands Lake, China.

The dotted horizontal line indicates the median of STI values; the dashed horizontal line indicates the mean of STI values. Grey points are Oriental species, red points are Palearctic species and purple points are Eurytopic species. The bottom right presents the frequency of STI values (right) and the overview relationship between total abundance (natural logarithm transversion of total abundance over 10 years and on all islands) and STI.



Appendix 1—figure 4.

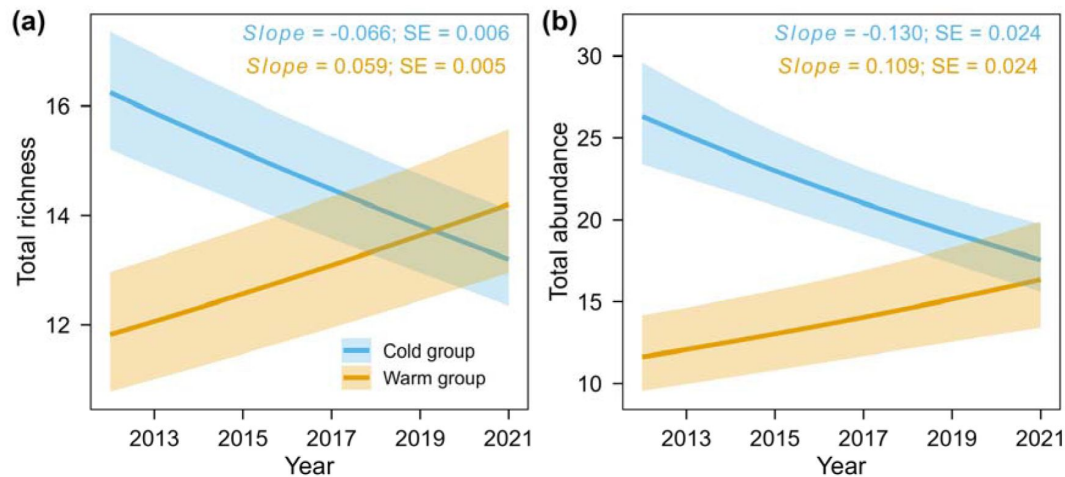
Relationship between STI and mean estimates and 95%, and 80% credible intervals of posterior distribution for temporal trends of species extinction rate (a) and species colonization rate (b).

Points and error bars in blue or orange indicate significant posterior estimates and error bars in grey indicate non-significant effects. Results of linear regression models modeling posterior mean of temporal trends in colonization or extinction weighted by standard deviations were added to plots.

Appendix 1—figure 5.

Partial regression effects of year effect on total species richness (a) and total abundance (b) for warm versus cold-adapted species groups.

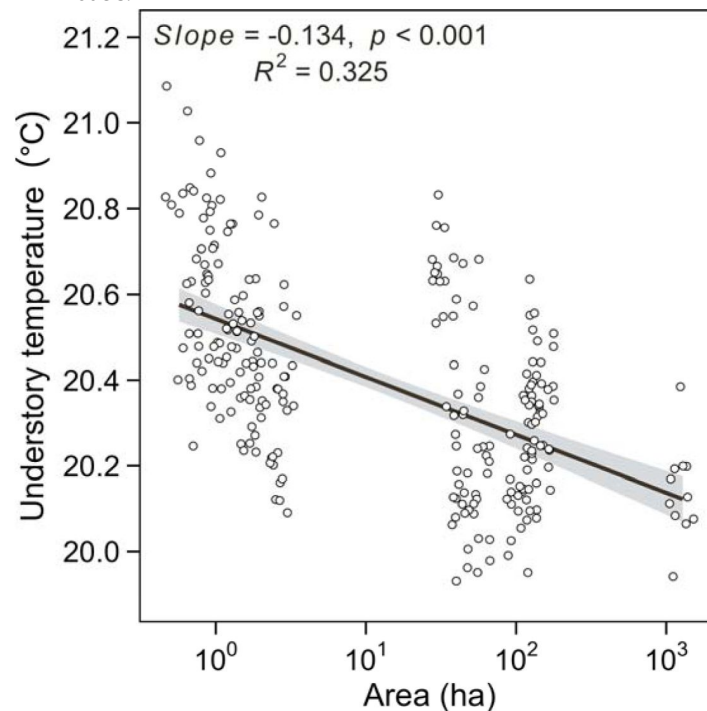
The solid lines indicate significant effects. The shaded bands are 95% confidence intervals of fitted values with other variables at their mean values. Total richness was calculated as the mean of the posterior estimate of species absence/presence, while total abundance was calculated from raw maximum abundance data.



Appendix 1—figure 6.

Relationship between island area and understory breeding season air temperature of 20 islands monitored in the year 2022.

The understory air temperature was monitored in 20 selected study islands differing in island area. Each plot represents the average air temperature (April – June) of a site within the islands. Lines and shaded areas are predicted values and 95% confidence intervals from LM models.



References

- Anderson C. M., Fahrig L., Rausch J., Martin J.-L., Daufresne T., Smith P. A (2023) **Climate-related range shifts in Arctic-breeding shorebirds** *Ecology and Evolution* **13** <https://doi.org/10.1002/ece3.9797>
- Barnagaud J. Y., Barbaro L., Hampe A., Jiguet F., Archaux F (2013) **Species' thermal preferences affect forest bird communities along landscape and local scale habitat gradients** *Ecography* **36**:1218–1226 <https://doi.org/10.1111/j.1600-0587.2012.00227.x>
- Bates A. E., Stuart-Smith R. D., Barrett N. S., Edgar G. J (2017) **Biological interactions both facilitate and resist climate-related functional change in temperate reef communities** *Proceedings of the Royal Society B: Biological Sciences* **284** <https://doi.org/10.1098/rspb.2017.0484>
- Bates D., Maechler M., Bolker B., Walker S (2015) **Fitting Linear Mixed-Effects Models Using lme4** *Journal of Statistical Software* **67**:1–48
- Brooker L., Brooker M., Cale P (1999) **Animal dispersal in fragmented habitat: measuring habitat connectivity, corridor use, and dispersal mortality** *Conservation Ecology* **3** <https://doi.org/10.5751/ES-00109-030104>
- Chen I.-C., Hill J. K., Ohlemüller R., Roy D. B., Thomas C. D (2011) **Rapid range shifts of species associated with high levels of climate warming** *Science* **333**:1024–1026 <https://doi.org/10.1126/science.1206432>
- Curley S. R., Manne L. L., Ramírez-Garofalo J. R., Veit R. R (2022) **Evaluating compositional changes in the avian communities of eastern North America using temperature and precipitation indices** *Journal of Biogeography* **49**:739–752 <https://doi.org/10.1111/jbi.14340>
- DeGregorio B. A., Westervelt J. D., Weatherhead P. J., Sperry J. H (2015) **Indirect effect of climate change: Shifts in ratsnake behavior alter intensity and timing of avian nest predation** *Ecological Modelling* **312**:239–246 <https://doi.org/10.1016/j.ecolmodel.2015.05.031>
- Devictor V., Julliard R., Couvet D., Jiguet F (2008) **Birds are tracking climate warming, but not fast enough** *Proceedings of the Royal Society B: Biological Sciences* **275**:2743–2748 <https://doi.org/10.1098/rspb.2008.0878>
- Dunn P. O., Møller A. P (2019) **Effects of climate change on birds** Oxford, UK: Oxford University Press
- Ewers R. M., Banks-Leite C (2013) **Fragmentation Impairs the Microclimate Buffering Effect of Tropical Forests** *PLOS ONE* **8** <https://doi.org/10.1371/journal.pone.0058093>
- Fahrig L (2003) **Effects of habitat fragmentation on biodiversity** *Annual review of ecology, evolution, and systematics* **34**:487–515
- Feeley K., Bravo-Avila C., Fadrique B., Perez T., Zuleta D (2020) **Climate-driven changes in the composition of New World plant communities** *Nature Climate Change* **10**:965–970 <https://doi.org/10.1038/s41558-020-0873-2>

- Fourcade Y., WallisDeVries M. F., Kuussaari M., van Swaay C. A., Heliölä J., Öckinger E. (2021) **Habitat amount and distribution modify community dynamics under climate change** *Ecology Letters* **24**:950–957 <https://doi.org/10.1111/ele.13691>
- Freeman B. G., Song Y., Feeley K. J., Zhu K (2021) **Montane species track rising temperatures better in the tropics than in the temperate zone** *Ecology Letters* **24**:1697–1708 <https://doi.org/10.1111/ele.13762>
- Gaüzère P., Princé K., Devictor V (2017) **Where do they go? The effects of topography and habitat diversity on reducing climatic debt in birds** *Global Change Biology* **23**:2218–2229 <https://doi.org/10.1111/gcb.13500>
- Gelman A., Meng X.-L., Stern H (1996) **Posterior predictive assessment of model fitness via realized discrepancies** *Statistica sinica* **6**:733–760 <https://doi.org/10.2307/24306036>
- Gelman A., Rubin D. B (1992) **Inference from iterative simulation using multiple sequences** *Statistical science* **7**:457–472 <https://doi.org/10.1214/ss/1177011136>
- Gottfried M., Pauli H., Futschik A., Akhalkatsi M., Barančok P., Alonso J. L. B., ..., Kazakis G. (2012) **Continent-wide response of mountain vegetation to climate change** *Nature Climate Change* **2**:111–115 <https://doi.org/10.1038/NCLIMATE1329>
- Harrison X. A (2014) **Using observation-level random effects to model overdispersion in count data in ecology and evolution** *PeerJ* **2** <https://doi.org/10.7717/peerj.616>
- Hu R., Gu Y., Luo M., Lu Z., Wei M., Zhong J (2020) **Shifts in bird ranges and conservation priorities in China under climate change** *PLOS ONE* **15** <https://doi.org/10.1371/journal.pone.0240225>
- Hylander K., Ehrlén J (2013) **The mechanisms causing extinction debts** *Trends in ecology & evolution* **28**:341–346 <https://doi.org/10.1016/j.tree.2013.01.010>
- La Sorte F. A., Jetz W. (2012) **Tracking of climatic niche boundaries under recent climate change** *Journal of Animal Ecology* **81**:914–925 <https://doi.org/10.1111/j.1365-2656.2012.01958.x>
- La Sorte F. A., Thompson F. R. (2007) **Poleward shifts in winter ranges of North American birds** *Ecology* **88**:1803–1812 <https://doi.org/10.1890/06-1072.1>
- Lajeunesse A., Fourcade Y (2022) **Temporal analysis of GBIF data reveals the restructuring of communities following climate change** *Journal of Animal Ecology* **92**:391–402 <https://doi.org/10.1111/1365-2656.13854>
- Lindström Å., Green M., Paulson G., Smith H. G., Devictor V. (2013) **Rapid changes in bird community composition at multiple temporal and spatial scales in response to recent climate change** *Ecography* **36**:313–322 <https://doi.org/10.1111/j.1600-0587.2012.07799.x>
- Liu J., Matthews T. J., Zhong L., Liu J., Wu D., Yu M (2020) **Environmental filtering underpins the island species—area relationship in a subtropical anthropogenic archipelago** *Journal of Ecology* **108**:424–432 <https://doi.org/10.1111/1365-2745.13272>
- Lüdecke D (2018) **ggeffects: Tidy data frames of marginal effects from regression models** *Journal of Open Source Software* **3** <https://doi.org/10.21105/joss.00772>

- Lüdecke D., Ben-Shachar M. S., Patil I., Waggoner P., Makowski D (2021) **performance: An R Package for Assessment Comparison and Testing of Statistical Models** *Journal of Open Source Software* **6**
- Luoto M., Heikkinen R (2008) **Disregarding topographical heterogeneity biases species turnover assessments based on bioclimatic models** *Global Change Biology* **14**:483–494 <https://doi.org/10.1111/j.1365-2486.2007.01527.x>
- MacKenzie D. I., Bailey L. L., Nichols J. D (2004) **Investigating species co-occurrence patterns when species are detected imperfectly** *Journal of Animal Ecology* **73**:546–555 <https://doi.org/10.1111/j.0021-8790.2004.00828.x>
- MacKenzie D. I., Nichols J. D., Hines J. E., Knutson M. G., Franklin A. B (2003) **Estimating site occupancy, colonization, and local extinction when a species is detected imperfectly** *Ecology* **84**:2200–2207 <https://doi.org/10.1890/02-3090>
- MacKenzie D. I., Nichols J. D., Royle J. A., Pollock K. H., Hines J. E (2006) **Occupancy estimation and modeling: inferring patterns and dynamics of species occurrence** Academic Press
- McGuire J. L., Lawler J. J., McRae B. H., Nuñez T. A., Theobald D. M (2016) **Achieving climate connectivity in a fragmented landscape** *Proceedings of the National Academy of Sciences* **113**:7195–7200 <https://doi.org/10.1073/pnas.1602817113>
- Menéndez R., Megías A. G., Hill J. K., Braschler B., Willis S. G., Collingham Y., ..., Thomas C. D (2006) **Species richness changes lag behind climate change** *Proceedings of the Royal Society B: Biological Sciences* **273**:1465–1470
- Nakagawa S., Schielzeth H (2013) **A general and simple method for obtaining R² from generalized linear mixed-effects models** *Methods in Ecology and Evolution* **4**:133–142 <https://doi.org/10.1111/j.2041-210x.2012.00261.x>
- Newson S. E., Oliver T. H., Gillings S., Crick H. Q., Morecroft M. D., Duffield S. J., ..., Pearce-Higgins J. W (2014) **Can site and landscape-scale environmental attributes buffer bird populations against weather events?** *Ecography* **37**:872–882 <https://doi.org/10.1111/ecog.00575>
- Oliver T. H., Gillings S., Pearce-Higgins J. W., Brereton T., Crick H. Q., Duffield S. J., ..., Roy D. B (2017) **Large extents of intensive land use limit community reorganization during climate warming** *Global Change Biology* **23**:2272–2283 <https://doi.org/10.1111/gcb.13587>
- Opdam P., Wascher D (2004) **Climate change meets habitat fragmentation: linking landscape and biogeographical scale levels in research and conservation** *Biological Conservation* **117**:285–297 <https://doi.org/10.1016/j.biocon.2003.12.008>
- Paquette A., Hargreaves A. L (2021) **Biotic interactions are more often important at species' warm versus cool range edges** *Ecology Letters* **24**:2427–2438 <https://doi.org/10.1111/ele.13864>
- Pearce-Higgins J. W., Eglinton S. M., Martay B., Chamberlain D. E (2015) **Drivers of climate change impacts on bird communities** *Journal of Animal Ecology* **84**:943–954 <https://doi.org/10.1111/1365-2656.12364>
- Pinheiro J., Bates D., DebRoy S., Sarkar D., Heisterkamp S., Van Willigen B., Maintainer R. (2017) **nlme: Linear and Nonlinear Mixed Effects Models** *R package version 3*:1–162

Pinheiro J., Bates D., Team R. C (2023) **nlme: Linear and Nonlinear Mixed Effects Models R package version 3**:1–162

Platts P. J., Mason S. C., Palmer G., Hill J. K., Oliver T. H., Powney G. D., ..., Thomas C. D (2019) **Habitat availability explains variation in climate-driven range shifts across multiple taxonomic groups** *Scientific Reports* **9**:1–10 <https://doi.org/10.1038/s41598-019-51582-2>

Plummer M (2003) **JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling** *Paper presented at the Proceedings of the 3rd international workshop on distributed statistical computing*

Plummer M (2023) **rjags: Bayesian graphical models using MCMC R package version** :4–14

Prince K., Zuckerberg B (2015) **Climate change in our backyards: the reshuffling of North America's winter bird communities** *Global Change Biology* **21**:572–585 <https://doi.org/10.1111/gcb.12740>

Richard B., Dupouey J., Corcket E., Alard D., Archaux F., Aubert M., ..., Macé S. (2021) **The climatic debt is growing in the understorey of temperate forests: Stand characteristics matter** *Global Ecology and Biogeography* **30**:1474–1487 <https://doi.org/10.1111/geb.13312>

Rigal S., Dakos V., Alonso H., Auniņš A., Benkő Z., Brotons L., ..., Del Moral J. C (2023) **Farmland practices are driving bird population decline across Europe** *Proceedings of the National Academy of Sciences* **120** <https://doi.org/10.1073/pnas.2216573120>

Royle J. A., Kéry M (2007) **A Bayesian state-space formulation of dynamic occupancy models** *Ecology* **88**:1813–1823 <https://doi.org/10.1890/06-0669.1>

Sagarin R. D., Gaines S. D (2002) **The 'abundant centre' distribution: to what extent is it a biogeographical rule?** *Ecology Letters* **5**:137–147 <https://doi.org/10.1046/j.1461-0248.2002.00297.x>

Schielzeth H (2010) **Simple means to improve the interpretability of regression coefficients** *Methods in Ecology and Evolution* **1**:103–113 <https://doi.org/10.1111/j.2041-210X.2010.00012.x>

Schoereder J. H., Galbiati C., Ribas C. R., Sobrinho T. G., Sperber C. F., DeSouza O., Lopes-Andrade C (2004) **Should we use proportional sampling for species-area studies?** *Journal of Biogeography* **31**:1219–1226 <https://doi.org/10.1111/j.1365-2699.2004.01113.x>

Si X., Cadotte M. W., Zeng D., Baselga A., Zhao Y., Li J., ..., Ding P. (2017) **Functional and phylogenetic structure of island bird communities** *Journal of Animal Ecology* **86**:532–542 <https://doi.org/10.1111/1365-2656.12650>

Si X., Cadotte M. W., Zhao Y., Zhou H., Zeng D., Li J., ..., Tingley M. W (2018) **The importance of accounting for imperfect detection when estimating functional and phylogenetic community structure** *Ecology* **99**:2103–2112 <https://doi.org/10.1002/ecs.2438>

Si X., Pimm S. L., Russell G. J., Ding P (2014) **Turnover of breeding bird communities on islands in an inundated lake** *Journal of Biogeography* **41**:2283–2292 <https://doi.org/10.1111/jbi.12379>

Su Y.-S., Yajima M (2021) **R2jags: Using R to Run 'JAGS' R package version 0.7-1**

- Suggitt A. J., Wilson R. J., Isaac N. J., Beale C. M., Auffret A. G., August T., ..., Fox R. (2018) **Extinction risk from climate change is reduced by microclimatic buffering** *Nature Climate Change* **8**:713–717 <https://doi.org/10.1038/s41558-018-0231-9>
- Tayleur C. M., Devictor V., Gaüzère P., Jonzén N., Smith H. G., Lindström Å (2016) **Regional variation in climate change winners and losers highlights the rapid loss of cold-dwelling species** *Diversity and Distributions* **22**:468–480 <https://doi.org/10.1111/ddi.12412>
- Team R. C (2023) **R: A Language and Environment for Statistical Computing** In. Vienna, Austria: R Foundation for Statistical Computing
- Thomas C. D., Franco A. M., Hill J. K (2006) **Range retractions and extinction in the face of climate warming** *Trends in Ecology & Evolution* **21**:415–416 <https://doi.org/10.1016/j.tree.2006.05.012>
- Thomas C. D., Lennon J. J (1999) **Birds extend their ranges northwards** *Nature* **399**:213–213 <https://doi.org/10.1038/20335>
- Tingley M. W., Beissinger S. R (2009) **Detecting range shifts from historical species occurrences: new perspectives on old data** *Trends in ecology & evolution* **24**:625–633 <https://doi.org/10.1016/j.tree.2009.05.009>
- Tingley M. W., Beissinger S. R (2013) **Cryptic loss of montane avian richness and high community turnover over 100 years** *Ecology* **94**:598–609 <https://doi.org/10.1890/12-0928.1>
- Warren M., Hill J., Thomas J., Asher J., Fox R., Huntley B., ..., Harding P. (2001) **Rapid responses of British butterflies to opposing forces of climate and habitat change** *Nature* **414**:65–69 <https://doi.org/10.1038/35102054>
- Zellweger F., Frenne P. D., Lenoir J., Vangansbeke P., Coomes D (2020) **Forest microclimate dynamics drive plant responses to warming** *Science* **368**:772–775 <https://doi.org/10.1126/science.aba6880>

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Reviewer #3 (Public review):

Summary:

Juan Liu et al. investigated the interplay between habitat fragmentation and climate-driven thermophilization in birds in an island system in China. They used extensive bird monitoring data (9 surveys per year per island) across 36 islands of varying size and isolation from the mainland covering 10 years. The authors use extensive modeling frameworks to test a general increase of the occurrence and abundance of warm-dwelling species and vice versa

for cold-dwelling species using the widely used Community Temperature Index (CTI), as well the relationship between island fragmentation in terms of island area and isolation from the mainland on extinction and colonization rates of cold- and warm-adapted species. They found that indeed there was thermophilization happening during the last 10 years, which was more pronounced for the CTI based on abundances and less clearly for the occurrence based metric. Generally, the authors show that this is driven by an increased colonization rate of warm-dwelling and an increased extinction rate of cold-dwelling species. Interestingly, they unravel some of the mechanisms behind this dynamic by showing that warm-adapted species increased while cold-dwelling decreased more strongly on smaller islands, which is - according to the authors - due to lowered thermal buffering on smaller islands (which was supported by air temperature monitoring done during the study period on small and large islands). They argue, that the increased extinction rate of cold-adapted species could also be due to lowered habitat heterogeneity on smaller islands. With regards to island isolation, they show that also both thermophilization processes (increase of warm and decrease of cold-adapted species) was stronger on islands closer to the mainland, due to closer sources to species populations of either group on the mainland as compared to limited dispersal (i.e. range shift potential) in more isolated islands.

The conclusions drawn in this study are sound, and mostly well supported by the results. Only few aspects leave open questions and could quite likely be further supported by the authors themselves thanks to their apparent extensive understanding of the study system.

Strengths:

The study questions and hypotheses are very well aligned with the methods used, ranging from field surveys to extensive modeling frameworks, as well as with the conclusions drawn from the results. The study addresses a complex question on the interplay between habitat fragmentation and climate-driven thermophilization which can naturally be affected by a multitude of additional factors than the ones included here. Nevertheless, the authors use a well balanced method of simplifying this to the most important factors in question (CTI change, extinction, colonization, together with habitat fragmentation metrics of isolation and island area). The interpretation of the results presents interesting mechanisms without being too bold on their findings and by providing important links to the existing literature as well as to additional data and analyses presented in the appendix.

Weaknesses:

The metric of island isolation based on distance to the mainland seems a bit too oversimplified as in real-life the study system rather represents an island network where the islands of different sizes are in varying distances to each other, such that smaller islands can potentially draw from the species pools from near-by larger islands too - rather than just from the mainland. Although the authors do explain the reason for this metric, backed up by earlier research, a network approach could be worthwhile exploring in future research done in this system. The fact, that the authors did find a signal of island isolation does support their method, but the variation in responses to this metric could hint on a more complex pattern going on in real-life than was assumed for this study.

<https://doi.org/10.7554/eLife.98056.2.sa1>

Author response:

The following is the authors' response to the original reviews.

We summarized the main changes:

- (1) In the Introduction part, we give a general definition of habitat fragmentation to avoid confusion, as reviewers #1 and #2 suggested.
- (2) We clarify the two aspects of the observed “extinction”——“true dieback” and “emigration”, as reviewers #2 and #3 suggested.
- (3) In the Methods part, we 1) clarify the reason for testing the temporal trend in colonization/extinction dynamics and describe how to select islands as reviewer #1 suggested; 2) describe how to exclude birds from the analysis as reviewer #2 suggested.
- (4) In the Results part, we modified and rearranged Figure 4-6 as reviewers #1, #2 and #3 suggested.
- (5) In the Discussion part, we 1) discuss the multiple aspects of the metric of isolation for future research as reviewer #3 suggested; 2) provide concrete evidence about the relationship between habitat diversity or heterogeneity and island area and 3) provide a wider perspective about how our results can inform conservation practices in fragmented habitats as reviewer #2 suggested.

eLife Assessment

This important study enhances our understanding of how habitat fragmentation and climate change jointly influence bird community thermophilization in a fragmented island system. The evidence supporting some conclusions is incomplete, as while the overall trends are convincing, some methodological aspects, particularly the isolation metrics and interpretation of colonization/extinction rates, require further clarification. This work will be of broad interest to ecologists and conservation biologists, providing crucial insights into how ecosystems and communities react to climate change.

We sincerely extend our gratitude to you and the esteemed reviewers for acknowledging the importance of our study and for raising these concerns. We have clarified the rationale behind our analysis of temporal trends in colonization and extinction dynamics, as well as the choice of distance to the mainland as the isolation metric. Additionally, we further discuss the multiple aspects of the metric of isolation for future research and provide concrete supporting evidence about the relationship between habitat diversity or heterogeneity and island area.

Incorporating these valuable suggestions, we have thoroughly revised our manuscript, ensuring that it now presents a more comprehensive and nuanced account of our research. We are confident that these improvements will further enhance the impact and relevance of our work for ecologists and conservation biologists alike, offering vital insights into the resilience and adaptation strategies of communities facing the challenges of climate change.

Reviewer #1 (Public Review):

Summary:

This study reports on the thermophilization of bird communities in a network of islands with varying areas and isolation in China. Using data from 10 years of transect surveys, the authors show that warm-adapted species tend to gradually replace cold-adapted species, both in terms of abundance and occurrence. The observed trends in colonisations and extinctions are related to the respective area and isolation of islands, showing an effect of fragmentation on the process of thermophilization.

Strengths:

Although thermophilization of bird communities has been already reported in different contexts, it is rare that this process can be related to habitat fragmentation, despite the fact that it has been hypothesized for a long time that it could play an important role. This is made possible thanks to a really nice study system in which the construction of a dam has created this incredible Thousand Islands lake. Here, authors do not simply take observed presence-absence as granted and instead develop an ambitious hierarchical dynamic multi-species occupancy model. Moreover, they carefully interpret their results in light of their knowledge of the ecology of the species involved.

Response: We greatly appreciate your recognition of our study system and the comprehensive approach and careful interpretation of results.

Weaknesses:

Despite the clarity of this paper on many aspects, I see a strong weakness in the authors' hypotheses, which obscures the interpretation of their results. Looking at Figure 1, and in many sentences of the text, a strong baseline hypothesis is that thermophilization occurs because of an increasing colonisation rate of warm-adapted species and extinction rate of cold-adapted species. However, there does not need to be a temporal trend! Any warm-adapted species that colonizes a site has a positive net effect on CTI; similarly, any cold-adapted species that goes extinct contributes to thermophilization.

Thank you very much for these thoughtful comments. The understanding depends on the time frame of the study and specifically, whether the system is at equilibrium. We think your claim is based on this background: if the system is not at equilibrium, then CTI can shift simply by having differential colonization (or extinction) rates for warm-adapted versus cold-adapted species. We agree with you in this case.

On the other hand, if a community is at equilibrium, then there will be no net change in CTI over time. Imagine we have an archipelago where the average colonization of warm-adapted species is larger than the average colonization of cold-adapted species, then over time the archipelago will reach an equilibrium with stable colonization/extinction dynamics where the average CTI is stable over time. Once it is stable, then if there is a temporal trend in colonization rates, the CTI will change until a new equilibrium is reached (if it is reached).

For our system, the question then is whether we can assume that the system is or has ever been at equilibrium. If it is not at equilibrium, then CTI can shift simply by having differential colonization (or extinction) rates for warm-adapted versus cold-adapted species. If the system is at equilibrium (at the beginning of the study), then CTI will only shift if there is a temporal change or trend in colonization or extinction rates.

Habitat fragmentation can affect biomes for decades after dam formation. The “Relaxation effect” (Gonzalez, 2000) refers to the fact that the continent acts as a potential species pool for island communities. Under relaxation, some species will be filtered out over time, mainly through the selective extinction of species that are highly sensitive to fragmentation. Meanwhile, for a 100-hectare patch, it takes about ten years to lose 50% of bird species; The smaller the patch area, the shorter the time required (Ferraz et al., 2003; Haddad et al., 2015). This study was conducted 50 to 60 years after the formation of the TIL, making the system with a high probability of reaching “equilibrium” through “Relaxation effect”(Si et al., 2014). We have no way of knowing exactly whether “equilibrium” is true in our system. Thus, changing rates of colonization-extinction over time is actually a much stronger test of thermophilization, which makes our inference more robust.

We add a note to the legend of Figure 1 on Lines 781-786:

“CTI can also change simply due to differential colonization-extinction rates by thermal affinity if the system is not at equilibrium prior to the study. In our study system, we have no way of knowing whether our island system was at equilibrium at onset of the study, thus, focusing on changing rates of colonization-extinction over time presents a much stronger tests of thermophilization.”

We hope this statement can make it clear. Thank you again for this meaningful question.

Another potential weakness is that fragmentation is not clearly defined. Generally, fragmentation sensu lato involves both loss of habitat area and changes in the spatial structure of habitats (i.e. fragmentation per se). Here, both area and isolation are considered, which may be slightly confusing for the readers if not properly defined.

Thank you for reminding us of that. Habitat fragmentation in this study involves both habitat loss and fragmentation per se. We have clarified the general definition in the Introduction on Lines 61-63:

“Habitat fragmentation, usually defined as the shifts of continuous habitat into spatially isolated and small patches (Fahrig, 2003), in particular, has been hypothesized to have interactive effects with climate change on community dynamics.”

Reviewer #2 (Public Review):

Summary:

This study addresses whether bird community reassembly in time is related to climate change by modelling a widely used metric, the community temperature index (CTI). The authors first computed the temperature index of 60 breeding bird species thanks to distribution atlases and climatic maps, thus obtaining a measure of the species realized thermal niche.

These indices were aggregated at the community level, using 53 survey transects of 36 islands (repeated for 10 years) of the Thousand Islands Lake, eastern China. Any increment of this CTI (i.e. thermophilization) can thus be interpreted as a community reassembly caused by a change in climate conditions (given no confounding correlations).

The authors show thanks to a mix of Bayesian and frequentist mixed effect models to study an increment of CTI at the island level, driven by both extinction (or emigration) of cold-adapted species and colonization of newly adapted warm-adapted species. Less isolated islands displayed higher colonization and extinction rates, confirming that dispersal constraints (created by habitat fragmentation per se) on colonization and emigration are the main determinants of thermophilization. The authors also had the opportunity to test for habitat amount (here island size). They show that the lack of microclimatic buffering resulting from less forest amount (a claim backed by understory temperature data) exacerbated the rates of cold-adapted species extinction while fostering the establishment of warm-adapted species.

Overall these findings are important to range studies as they reveal the local change in affinity to the climate of species comprising communities while showing that the habitat fragmentation VS amount distinction is relevant when studying thermophilization. As is, the manuscript lacks a wider perspective about how these results can be fed into conservation biology, but would greatly benefit from it. Indeed, this study shows that in a fragmented reserve context, habitat amount is very important in explaining trends of loss of cold-adapted species, hinting that it may be strategic to prioritize large habitats to conserve such species. Areas of diverse size may act as stepping stones for species

shifting range due to climate change, with small islands fostering the establishment of newly adapted warm-adapted species while large islands act as refugia for cold-adapted species. This study also shows that the removal of dispersal constraints with low isolation may help species relocate to the best suitable microclimate in a heterogenous reserve context.

Thank you very much for your valuable feedback. We greatly appreciate your recognition of the scientific question to the extensive dataset and diverse approach. In particular, you provided constructive suggestions and examples on how to extend the results to conservation guidance. This is something we can't ignore in the manuscript. We have added a paragraph to the end of the Discussion, stating how our results can inform conservation, on Lines 339-347:

‘Overall, our findings have important implications for conservation practices. Firstly, we confirmed the role of isolation in limiting range shifting. Better connected landscapes should be developed to remove dispersal constraints and facilitate species’ relocation to the best suitable microclimate. Second, small patches can foster the establishment of newly adapted warm-adapted species while large patches can act as refugia for cold-adapted species. Therefore, preserving patches of diverse sizes can act as stepping stones or shelters in a warming climate depending on the thermal affinity of species. These insights are important supplement to the previous emphasis on the role of habitat diversity in fostering (Richard et al., 2021) or reducing (Gaüzère et al., 2017) community-level climate debt.’

Strength:

The strength of the study lies in its impressive dataset of bird resurveys, that cover 10 years of continued warming (as evidenced by weather data), 60 species in 36 islands of varying size and isolation, perfect for disentangling habitat fragmentation and habitat amount effects on communities. This distinction allows us to test very different processes mediating thermophilization; island area, linked to microclimatic buffering, explained rates for a variety of species. Dispersal constraints due to fragmentation were harder to detect but confirms that fragmentation does slow down thermophilization processes.

This study is a very good example of how the expected range shift at the biome scale of the species materializes in small fragmented regions. Specifically, the regional dynamics the authors show are analogous to what processes are expected at the trailing and colonizing edge of a shifting range: warmer and more connected places display the fastest turnover rates of community reassembly. The authors also successfully estimated extinction and colonization rates, allowing a more mechanistic understanding of CTI increment, being the product of two processes.

The authors showed that regional diversity and CTI computed only by occurrences do not respond in 10 years of warming, but that finer metrics (abundance-based, or individual islands considered) do respond. This highlights the need to consider a variety of case-specific metrics to address local or regional trends. Figure Appendix 2 is a much-appreciated visualization of the effect of different data sources on Species thermal Index (STI) calculation.

The methods are long and diverse, but they are documented enough so that an experienced user with the use of the provided R script can follow and reproduce them.

Thank you very much for your profound Public Review. We greatly appreciate your recognition of the scientific question, the extensive dataset and the diverse approach.

Weaknesses:

While the overall message of the paper is supported by data, the claims are not uniformly backed by the analysis. The trends of island-specific thermophilization are very credible (Figure 3), however, the variable nature of bird observations (partly compensated by an impressive number of resurveys) propagate a lot of errors in the estimation of species-specific trends in occupancy, abundance change, and the extinction and colonization rates. This materializes into a weak relationship between STI and their respective occupancy and abundance change trends (Figure 4a, Figure 5, respectively), showing that species do not uniformly contribute to the trend observed in Figure 3. This is further shown by the results presented in Figure 6, which present in my opinion the topical finding of the study. While a lot of species rates response to island areas are significant, the isolation effect on colonization and extinction rates can only be interpreted as a trend as only a few species have a significant effect. The actual effect on the occupancy change rates of species is hard to grasp, and this trend has a potentially low magnitude (see below).

Thank you very much for pointing out this shortcoming. The R^2 between STI and their respective occupancy trends is relatively small ($R^2=0.035$). But the R^2 between STI and their respective abundance change trends are relatively bigger, in the context of Ecology research ($R^2=0.123$). The R^2 between STI and their respective colonization rate ($R^2=0.083$) and extinction rate trends ($R^2=0.053$) are also relatively small. Low R^2 indicates that we can't make predictions using the current model, we must notice that except STI, other factors may influence the species-specific occupancy trend. Nonetheless, it is important to notice that the standardized coefficient estimates are not minor and the trend is also significant, indicating the species-specific response is at least related to STI.

The number of species that have significant interaction terms for isolation (Figure 6) is indeed low. Although there is uncertainty in the estimation of relationships, there are also consistent trends in response to habitat fragmentation of colonization of warm-adapted species and extinction of cold-adapted species. This is especially true for the effect of isolation, where on islands nearer to the mainland, warm-adapted species (15 out of 15 investigated species) increased their colonization probability at a higher rate over time, while most cold-adapted species (21 out of 23 species) increased their extinction probability at a higher rate. We now better highlight these results in the Results and Discussion.

While being well documented, the myriad of statistical methods used by the authors ampere the interpretation of the figure as the posterior mean presented in Figure 4b and Figure 6 needs to be transformed again by a logit-1 and fed into the equation of the respective model to make sense of. I suggest a rewording of the caption to limit its dependence on the method section for interpretation.

Thank you for this suggestion. The value on the Y axis indicates the posterior mean of each variable (year, area, isolation and their interaction effects) extracted from the MSOM model, where the $\text{logit}(\text{extinction rate})$ or $\text{logit}(\text{colonization rate})$ was the response variable. All variables were standardized before analysis to make them comparable so interpretation is actually quite straight forward: positive values indicate positive influence while negative values indicate negative influence. Because the goal of Figure 6 is to display the negative/positive effect, we didn't back-transform them. Following your advice, we thus modified the caption of Figure 6 (now renumbered as Figure 5, following a comment from Reviewer #3, to move Figure 5 to Figure 4c). The modified title and legends of Figure 5 are on Lines 817-820:

“Figure 5. Posterior estimates of logit-scale parameters related to cold-adapted species' extinction rates and warm-adapted species' colonization rates. Points are species-specific

posterior means on the logit-scale, where parameters >0 indicate positive effects (on extinction [a] or colonization [b]) and parameters <0 indicate negative effects...”

By using a broad estimate of the realized thermal niche, a common weakness of thermophilization studies is the inability to capture local adaptation in species' physiological or behavioral response to a rise in temperature. The authors however acknowledge this limitation and provide specific examples of how species ought to evade high temperatures in this study region.

We appreciate your recognition. This is a common problem in STI studies. We hope in future studies, researchers can take more details about microclimate of species' true habitat across regions into consideration when calculating STI. Although challenging, focusing on a smaller portion of its distribution range may facilitate achievement.

Reviewer #3 (Public Review):

Summary:

Juan Liu et al. investigated the interplay between habitat fragmentation and climate-driven thermophilization in birds in an island system in China. They used extensive bird monitoring data (9 surveys per year per island) across 36 islands of varying size and isolation from the mainland covering 10 years. The authors use extensive modeling frameworks to test a general increase in the occurrence and abundance of warm-dwelling species and vice versa for cold-dwelling species using the widely used Community Temperature Index (CTI), as well as the relationship between island fragmentation in terms of island area and isolation from the mainland on extinction and colonization rates of cold- and warm-adapted species. They found that indeed there was thermophilization happening during the last 10 years, which was more pronounced for the CTI based on abundances and less clearly for the occurrence-based metric. Generally, the authors show that this is driven by an increased colonization rate of warm-dwelling and an increased extinction rate of cold-dwelling species. Interestingly, they unravel some of the mechanisms behind this dynamic by showing that warm-adapted species increased while cold-dwelling decreased more strongly on smaller islands, which is - according to the authors - due to lowered thermal buffering on smaller islands (which was supported by air temperature monitoring done during the study period on small and large islands). They argue, that the increased extinction rate of cold-adapted species could also be due to lowered habitat heterogeneity on smaller islands. With regards to island isolation, they show that also both thermophilization processes (increase of warm and decrease of cold-adapted species) were stronger on islands closer to the mainland, due to closer sources to species populations of either group on the mainland as compared to limited dispersal (i.e. range shift potential) in more isolated islands.

The conclusions drawn in this study are sound, and mostly well supported by the results. Only a few aspects leave open questions and could quite likely be further supported by the authors themselves thanks to their apparent extensive understanding of the study system.

Strengths:

The study questions and hypotheses are very well aligned with the methods used, ranging from field surveys to extensive modeling frameworks, as well as with the conclusions drawn from the results. The study addresses a complex question on the interplay between habitat fragmentation and climate-driven thermophilization which can naturally be affected by a multitude of additional factors than the ones included here. Nevertheless, the authors use a well-balanced method of simplifying this to the most important factors in question (CTI change, extinction, and colonization, together with

habitat fragmentation metrics of isolation and island area). The interpretation of the results presents interesting mechanisms without being too bold on their findings and by providing important links to the existing literature as well as to additional data and analyses presented in the appendix.

We appreciate very much for your positive and constructive comments and suggestions. Thank you for your recognition of the scientific question, the modeling approach and the conclusions.

Weaknesses:

The metric of island isolation based on the distance to the mainland seems a bit too oversimplified as in real life the study system rather represents an island network where the islands of different sizes are in varying distances to each other, such that smaller islands can potentially draw from the species pools from near-by larger islands too - rather than just from the mainland. Thus a more holistic network metric of isolation could have been applied or at least discussed for future research. The fact, that the authors did find a signal of island isolation does support their method, but the variation in responses to this metric could hint at a more complex pattern going on in real-life than was assumed for this study.

Thank you for this meaningful question. Isolation can be measured in different ways in the study region. We chose the distance to the mainland as a measure of isolation based on the results of a previous study. One study in our system provided evidence that the colonization rate and extinction rate of breeding bird species were best fitted using distance to the nearest mainland over other distance-based measures (distance to the nearest landmass, distance to the nearest bigger landmass)(Si et al., 2014). Besides, their results produced almost identical patterns of the relationship between isolation and colonization/extinction rate (Si et al., 2014). That's why we only selected "Distance to the mainland" in our current analysis and we do find some consistent patterns as expected. The plants on all islands were cleared out about 60 years ago due to dam construction, with all bird species coming from the mainland as the original species pool through a process called "relaxation". This could be the reason why distance to the nearest mainland is the best predictor.

We agree with you that it's still necessary to consider more aspects of "isolation" at least in discussion for future research. In our Discussion, we address these on Lines 292-299:

"As a caveat, we only consider the distance to the nearest mainland as a measure of fragmentation, consistent with previous work in this system (Si et al., 2014), but we acknowledge that other distance-based metrics of isolation that incorporate inter-island connections could reveal additional insights on fragmentation effects. The spatial arrangement of islands, like the arrangement of habitat, can influence niche tracking of species (Fourcade et al., 2021). Future studies should take these metrics into account to thoroughly understand the influence of isolation and spatial arrangement of patches in mediating the effect of climate warming on species."

Further, the link between larger areas and higher habitat diversity or heterogeneity could be presented by providing evidence for this relationship. The authors do make a reference to a paper done in the same study system, but a more thorough presentation of it would strengthen this assumption further.

Thank you very much for this question. We now add more details about the relationship between habitat diversity and heterogeneity based on a related study in the same system. The observed number of species significantly increased with increasing island area (slope = 4.42, $R^2 = 0.70$, $p < .001$), as did the rarefied species richness per island (slope = 1.03, $R^2 = 0.43$, $p <$

.001), species density (slope = 0.80, $R^2 = 0.33$, $p = .001$) and the rarefied species richness per unit area (slope = 0.321, $R^2 = 0.32$, $p = .001$). We added this supporting evidence on Lines 317-321:

“We thus suppose that habitat heterogeneity could also mitigate the loss of these relatively cold-adapted species as expected. Habitat diversity, including the observed number of species, the rarefied species richness per island, species density and the rarefied species richness per unit area, all increased significantly with island area instead of isolation in our system (Liu et al., 2020)”

Despite the general clear patterns found in the paper, there were some idiosyncratic responses. Those could be due to a multitude of factors which could be discussed a bit better to inform future research using a similar study design.

Thank you for these suggestions. We added a summary statement about the reasons for idiosyncratic responses on Lines 334-338:

“Overall, these idiosyncratic responses reveal several possible mechanisms in regulating species' climate responses, including resource demands and biological interactions like competition and predation. Future studies are needed to take these factors into account to understand the complex mechanisms by which habitat loss mediates species range shifts.”

Reviewer #1 (Recommendations For The Authors):

(1) Figure 1: I disagree that there should be a temporal trend in colonisation/extinction dynamics.

Thank you again for these thoughtful comments. We have explained in detail in the response to the Public Review.

(2) L 485-487: As explained before I disagree. I don't see why there needs to be a temporal trend in colonization and extinction.

Thank you again for these thoughtful comments. Because we can't guarantee that the study system has reached equilibrium, changing rates of colonization-extinction over time is actually a much stronger test of thermophilization. More detailed statement can be seen in the response to the Public Review.

(3) L 141: which species' ecological traits?

Sorry for the confusion. The traits included continuous variables (dispersal ability, body size, body mass and clutch size) and categorical variables (diet, active layer, residence type). Specifically, we tested the correlation between STI and dispersal ability, body size, body mass and clutch size using Pearson correlation test. We also tested the difference in STI between different trait groups using the Wilcoxon signed-rank test for three Category variables: diet (carnivorous/ omnivorous/ herbivory), active layer (canopy/mid/low), and residence type (resident species/summer visitor). There is no significant difference between any two groups for each of the three category variables ($p > 0.2$). We added these on Lines 141-145:

“No significant correlation was found between STI and species' ecological traits; specifically, the continuous variables of dispersal ability, body size, body mass and clutch size (Pearson correlations for each, $|r| < 0.22$), and the categorical variables of diet (carnivorous/omnivorous/herbivory), active layer (canopy/mid/low), and residence type (resident species/summer visitor)”

(4) L 143: *CTIoccur and CTIabun were not defined before.*

Because CTIoccur and CTIabun were first defined in Methods part (section 4.4), we change the sentence to a more general statement here on Lines 147-150:

“At the landscape scale, considering species detected across the study area, occurrence-based CTI (CTIoccur; see section 4.4) showed no trend (posterior mean temporal trend = 0.414; 95% CrI: -12.751, 13.554) but abundance-based CTI (CTIabun; see section 4.4) showed a significant increasing trend.”

(5) *Figure 4: what is the dashed vertical line? I assume the mean STI across species?*

Sorry for the unclear description. The vertical dashed line indicates the median value of STI for 60 species, as a separation of warm-adapted species and cold-adapted species. We have added these details on Lines 807-809:

“The dotted vertical line indicates the median of STI values. Cold-adapted species are plotted in blue and warm-adapted species are plotted in orange.”

(6) *Figure 6: in the legend, replace 'points in blue' with 'points in blue/orange' or 'solid dots' or something similar.*

Thank you for this suggestion. We changed it to “points in blue/orange” on Lines 823.

(7) L 176-176: *unclear why the interaction parameters are particularly important for explaining the thermophilization mechanism: if e.g. colonization rate of warm-adapted species is constantly higher in less isolated islands, (and always higher than the extinction rate of the same species), it means that thermophilization is increased in less isolated islands, right?*

Thank you for this question. This is also related to the question about “Why use temporal trends in colonization/extinction rate to test for thermophilization mechanisms”. Colonization-extinction over time is actually a much stronger test of thermophilization (more details refer to response to Public Review and Recommendations 1&2).

Based on this, the two main driving processes of thermophilization mechanism include the increasing colonization rate of warm-adapted species and the increasing extinction rate of cold-adapted species with year. The interaction effect between island area (or isolation) and year on colonization rate (or extinction rate) can tell us how habitat fragmentation mediates the year effect. For example, if the interaction term between year and isolation is negative for a warm-adapted species that increased in colonization rate with year, it indicates that the colonization rate increased faster on less isolated islands. This is a signal of a faster thermophilization rate on less-isolated islands.

(8) L201-203: *this is only little supported by the results that actually show that there is NO significant interaction for most species.*

Thank you for this comment. Although most species showed non-significant interaction effect, the overall trend is relatively consistent, this is especially true for the effect of isolation. To emphasize the “trend” instead of “significant effect”, we slightly modified this sentence in more rigorous wording on Lines 205-208:

“We further found that habitat fragmentation influences two processes of thermophilization: colonization rates of most warm-adapted species tended to increase faster on smaller and less

isolated islands, while the loss rates of most cold-adapted species tended to be exacerbated on less isolated islands.”

(9) Section 2.3: *can't you have a population-level estimate? I struggled a bit to understand all the parameters of the MSOM (because of my lack of statistical/mathematical proficiency) so I cannot provide more advice here.*

Thank you for raising this advice. We think what you are mentioning is the overall estimate across all species for each variable. From MSOM, we can get a standardized estimate of every variable (year, area, isolation, interaction) for each species, separately. Because the divergent or consistent responses among species are what we are interested in, we didn't calculate further to get a population-level estimate.

(10) L 291: *a dot is missing.*

Done. Thank you for your correction.

(11) L 305, 315: *a space is missing*

Done

(12) L 332: *how were these islands selected?*

Thank you for this question. The 36 islands were selected according to a gradient of island area and isolation, spreading across the whole lake region. The selected islands guaranteed there is no significant correlation between island area and isolation (the Pearson correlation coefficient $r = -0.21$, $p = 0.21$). The biggest 7 islands among the 36 islands are also the only several islands larger than 30 ha in the whole lake region. We have modified this in the Method part on Lines 360-363.

“We selected 36 islands according to a gradient of island area and isolation with a guarantee of no significant correlation between island area and isolation (Pearson $r = -0.21$, $p = 0.21$). For each island, we calculated island area and isolation (measured in the nearest Euclidean distance to the mainland) to represent the degree of habitat fragmentation.”

(13) L 334: *"Distance to the mainland" was used as a metric of isolation, but elsewhere in the text you argue that the observed thermophilization is due to interisland movements. It sounds contradictory. Why not include the average or shortest distance to the other islands?*

Thank you very much for raising this comment. Yes, “Distance to the mainland” was the only metric we used for isolation. We carefully checked through the manuscript where the “interisland movement” comes from and induces the misunderstanding. It must come from Discussion 3.1 (n Lines 217-221): “Notably, when tested on the landscape scale (versus on individual island communities), only the abundance-based thermophilization trend was significant, indicating thermophilization of bird communities was mostly due to inter-island occurrence dynamics, rather than exogenous community turnover.”

Sorry, the word “inter-island” is not exactly what we want to express here, we wanted to express that “the thermophilization was mostly due to occurrence dynamics within the region, rather than exogenous community turnover outside the region”. We have changed the sentence in Discussion part on Lines 217-221:

“Notably, when tested on the landscape scale (versus on individual island communities), only the abundance-based thermophilization trend was significant, indicating thermophilization

of bird communities was mostly due to occurrence dynamics within the region, rather than exogenous community turnover outside the region.”

Besides, I would like to explain why we use distance to the mainland. We chose the distance to the mainland as a measure of isolation based on the results of a previous study. One study in our system provided evidence that the colonization rate and extinction rate of breeding bird species were best fitted using distance to the nearest mainland over other distance-based measures (distance to the nearest landmass, distance to the nearest bigger landmass)(Si et al., 2014). Besides, their results produced almost identical patterns of the relationship between isolation and colonization/extinction rate(Si et al., 2014). That’s why we only selected “Distance to the mainland” in our current analysis and we do find some consistent patterns as expected. The plants on all islands were cleared out about 60 years ago due to dam construction, with all bird species coming from the mainland as the original species pool through a process called “relaxation”. This may be the reason why distance to the nearest mainland is the best predictor.

In Discussion part, we added the following discussion and talked about the other measures on Lines 292-299:

“As a caveat, we only consider the distance to the nearest mainland as a measure of fragmentation, consistent with previous work in this system (Si et al., 2014), but we acknowledge that other distance-based metrics of isolation that incorporate inter-island connections could reveal additional insights on fragmentation effects. The spatial arrangement of islands, like the arrangement of habitat, can influence niche tracking of species (Fourcade et al., 2021). Future studies should take these metrics into account to thoroughly understand the influence of isolation and spatial arrangement of patches in mediating the effect of climate warming on species.”

(14) L 347: you write 'relative' abundance but this measure is not relative to anything. Better write something like "we based our abundance estimate on the maximum number of individuals recorded across the nine annual surveys".

Thank you for this suggestion, we have changed the sentence on Lines 377-379:

“We based our abundance estimate on the maximum number of individuals recorded across the nine annual surveys.”

(15) L 378: shouldn't the formula for CTI_{occur} be (equation in latex format):

$$CTI_{occur, j, t} = \frac{\sum_{i=1}^{N_{j,t}} STI_i}{N_{j,t}}$$

Where $N_{j,t}$ is the total number of species surveyed in the community j in year t

Thank you very much for this careful check, we have revised it on Lines 415, 417:

$$CTI_{occur, j, t} = \frac{\sum_{i=1}^{N_{j,t}} STI_i}{N_{j,t}}$$

“where $N_{j,t}$ is the total number of species surveyed in the community j in year t .”

Reviewer #2 (Recommendations For The Authors):

(1) Line 76: "weakly"

Done. Thank you for your correction.

(2) Line 98: I suggest a change to this sentence: "For example, habitat fragmentation renders habitats to be too isolated to be colonized, causing sedentary butterflies to lag more behind climate warming in Britain than mobile ones"

Thank you for this modification, we have changed it on Lines 99-101.

(3) Line 101: remove either "higher" or "increasing"

Done, we have removed "higher". Thank you for this advice.

(4) Line 102: "benefiting from near source of"

Done.

(5) Line 104: "emigrate"

Done.

(6) Introduction: I suggest making it more explicit what process you describe under the word "extinction". At first read, I thought you were only referring to the dieback of individuals, but you also included emigration as an extinction process. It also needs to be reworded in Fig 1 caption.

Thank you for this suggestion. Yes, we can't distinguish in our system between local extinction and emigration. The observed "extinction" of cold-adapted species over 10 years may involve two processes that usually occur in order: first "emigration" and then if can't emigrate or withstand, "real local dieback". It should also be included in the legend of Figure 1, as you said. We have modified the legend in Lines 780-781:

"Note that extinction here may include both the emigration of species and then the local extinction of species."

There is also one part in the Discussion that mentions this on Lines 287-291: "While we cannot truly distinguish in our system between local extinction and emigration, we suspect that given two islands equal except in isolation, and if both lose suitability due to climate change, individuals can easily emigrate from the island nearer to the mainland, while individuals on the more isolated island would be more likely to be trapped in place until the species went locally extinct due to a lack of rescue".

(7) I also suggest differentiating habitat fragmentation (distances between islands) and habitat amount (area) as explained in Fahrig 2013 (Rethinking patch size and isolation effects: the habitat amount hypothesis) and her latter paper. This will help the reader what lies behind the general trend of fragmentation: fragmentation per se and habitat amount reduction.

Thank you for this suggestion! Habitat fragmentation in this study involves both habitat loss and fragmentation per se. We now give a general definition of habitat fragmentation on Lines 61-63:

"Habitat fragmentation, usually defined as the shifts of continuous habitat into spatially isolated and small patches (Fahrig, 2003), in particular, has been hypothesized to have interactive effects with climate change on community dynamics."

(8) Line 136: is the "+-" refers to the standard deviation or confidence interval, I suggest being explicit about it once at the start of the results.

Thank you for reminding this. The "+-" refers to the standard deviation (SD). The modified sentence is now on Lines 135-139:

"The number of species detected in surveys on each island across the study period averaged 13.37 ± 6.26 (mean $\pm$ SD) species, ranging from 2 to 40 species, with an observed gamma diversity of 60 species. The STI of all 60 birds averaged 19.94 ± 3.58 °C (mean $\pm$ SD) and ranged from 9.30 °C (*Cuculus canorus*) to 27.20 °C (*Prinia inornata*), with a median of STI is 20.63 °C (Appendix 1—figure 2; Appendix 1—figure 3)."

(9) Line 143: please specify the unit of thermophilization.

The unit of thermophilization rate is the change in degree per unit year. Because in all analyses, predictor variables were z-transformed to make their effect comparable. We have added on Line 151:

"When measuring CTI trends for individual islands (expressed as °/ unit year)"

(10) Line 289: check if no word is missing from the sentence.

The sentence is: "In our study, a large proportion (11 out of 15) of warm-adapted species increasing in colonization rate and half (12 out of 23) of cold-adapted species increasing in extinction rate were changing more rapidly on smaller islands."

Given that we have defined the species that were included in testing the third prediction in both Methods part and Result part: 15 warm-adapted species that increased in colonization rate and 23 cold-adapted species that increased in extinction rate. We now remove this redundant information and rewrote the sentence as below on Lines 300-302:

"In our study, the colonization rate of a large proportion of warm-adapted species (11 out of 15) and the extinction rate of half of old-adapted species (12 out of 23) were increasing more rapidly on smaller islands."

(11) Line 319: I really miss a concluding statement of your discussion, your results are truly interesting and deserve to be summarized in two or three sentences, and maybe a perspective about how it can inform conservation practices in fragmented settings.

Thank you for this profound suggestion both in Public Review and here. We have added a paragraph to the end of the Discussion, stating how our results can inform conservation, on Lines 339-347:

"Overall, our findings have important implications for conservation practices. Firstly, we confirmed the role of isolation in limiting range shifting. Better connected landscapes should be developed to remove dispersal constraints and facilitate species' relocation to the best suitable microclimate. Second, small patches can foster the establishment of newly adapted warm-adapted species while large patches can act as refugia for cold-adapted species. Therefore, preserving patches of diverse sizes can act as stepping stones or shelters in a warming climate depending on the thermal affinity of species. These insights are important supplement to the previous emphasis on the role of habitat diversity in fostering (Richard et al., 2021) or reducing (Gaüzère et al., 2017) community-level climate debt."

(12) Line 335: I suggest " ... the islands has been protected by forbidding logging, ..."

Thanks for this wonderful suggestion. Done. The new sentence is now on Lines 365-366:

“Since lake formation, the islands have been protected by forbidding logging, allowing natural succession pathways to occur.”

| (13) Line 345: *this speed is unusually high for walking, check the speed.*

Sorry for the carelessness, it should be 2.0 km/h. It has been corrected on Lines 375-376:

“In each survey, observers walked along each transect at a constant speed (2.0 km/h) and recorded all the birds seen or heard on the survey islands.”

| (14) Line 351: *you could add a sentence explaining why that choice of species exclusion was made. Was made from the start of the monitoring program or did you exclude species afterward?*

We excluded them afterward. We excluded non-breeding species, nocturnal and crepuscular species, high-flying species passing over the islands (e.g., raptors, swallows) and strongly water-associated birds (e.g., cormorants). These records were recorded during monitoring, including some of them being on the shore of the island or high-flying above the island, and some nocturnal species were just spotted by accident.

We described more details about how to exclude species on Lines 379-387:

“We excluded non-breeding species, nocturnal and crepuscular species, high-flying species passing over the islands (e.g., raptors, swallows) and strongly water-associated birds (e.g., cormorants) from our record. First, our surveys were conducted during the day, so some nocturnal and crepuscular species, such as the owls and nightjars were excluded for inadequate survey design. Second, wagtail, kingfisher, and water birds such as ducks and herons were excluded because we were only interested in forest birds. Third, birds like swallows, and eagles who were usually flying or soaring in the air rather than staying on islands, were also excluded as it was difficult to determine their definite belonging islands. Following these operations, 60 species were finally retained.”

| (15) Line 370: *I suggest adding the range and median of STI.*

Thanks for this good suggestion. The range, mean $\pm$ SD of STI were already in the Results part, we added the median of STI there as well. The new sentence is now in Results part on Lines 137-139:

“The STI of all 60 birds averaged 19.94 ± 3.58 °C (mean $\pm$ SD) and ranged from 9.30 °C (*Cuculus canorus*) to 27.20 °C (*Prinia inornata*), with a median of 20.63 °C (Appendix 1—figure 2; Appendix 1—figure 3).”

| (16) Figure 4.b: *Is it possible to be more explicit about what that trend is? the coefficient of the regression Logit(ext/col) ~ year + ?*

Thank you for this advice. Your understanding is right: we can interpret it as the coefficient of the ‘year’ effect in the model. More specifically, the ‘year’ effect or temporal trend here is the ‘posterior mean’ of the posterior distribution of ‘year’ in the MSOM (Multi-species Occupancy Model), in the context of the Bayesian framework. We modified this sentence on Lines 811-813:

“ Each point in (b) represents the posterior mean estimate of year in colonization, extinction or occupancy rate for each species.”

(17) Figure 6: is it possible to provide an easily understandable meaning of the prior presented in the Y axis? E.g. "2 corresponds to a 90% probability for a species to go extinct at $T+1$ ", if not, please specify that it is the logit of a probability.

Thank you for this question both in Public Review and here. The value on the Y axis indicates the posterior mean of each variable (year, area, isolation and their interaction effects) extracted from the MSOM model, where the logit(extinction rate) or logit(colonization rate) was the response variable. All variables were standardized before analysis to make them comparable. So, positive values indicate positive influence while negative values indicate negative influence. Because the goal of Figure 6 is to display the negative/positive effect, we didn't back-transform them. Following your advice, we thus modified the caption of Figure 6 (now renumbered as Figure 5, following a comment from Reviewer #3, to move Figure 5 to Figure 4c). The modified title and legends of Figure 5 are on Lines 817-820:

"Figure 5. Posterior estimates of logit-scale parameters related to cold-adapted species' extinction rates and warm-adapted species' colonization rates. Points are species-specific posterior means on the logit-scale, where parameters >0 indicate positive effects (on extinction [a] or colonization [b]) and parameters <0 indicate negative effects."

(18) Line 773: points in blue only are significant? I suggest "points in color".

Thank you for your reminder. Points in blue and orange are all significant. We have revised the sentence on Line 823:

"Points in blue/orange indicate significant effects."

These are all small suggestions that may help you improve the readability of the final manuscript. I warmly thank you for the opportunity to review this impressive study.

We appreciate your careful review and profound suggestions. We believe these modifications will improve the final manuscript.

Reviewer #3 (Recommendations For The Authors):

I have a few minor suggestions for paper revision for your otherwise excellent manuscript. I wish to emphasize that it was a pleasure to read the manuscript and that I especially enjoyed a very nice flow throughout the ms from a nicely rounded introduction that led well into the research questions and hypotheses all the way to a good and solid discussion.

Thank you very much for your review and recognition. We have carefully checked all recommendations and addressed them in the manuscript.

(1) L 63: space before the bracket missing and I suggest moving the reference to the end of the sentence (directly after habitat fragmentation does not seem to make sense).

Thank you very much for this suggestion. The missed space was added, and the reference has been moved to the end of the sentence. We also add a general definition of habitat fragmentation. The new sentence is on Lines 61-64:

"Habitat fragmentation, usually defined as the shifts of continuous habitat into spatially isolated and small patches (Fahrig, 2003), in particular, has been hypothesized to have interactive effects with climate change on community dynamics."

| (2) L 102: I suggest to write "benefitting ..." instead.

Done.

| (3) L 103: higher extinction rates (add "s").

Done.

| (4) L 104: this should probably say "emigrate" and "climate warming".

Done.

| (5) L 130-133: this is true for emigration (more isolated islands show slower emigration). But what about increased local extinction, especially for small and isolated islands? Especially since you mentioned later in the manuscript that often emigration and extinction are difficult to identify or differentiate. Might be worth a thought here or somewhere in the discussion?

Thank you for this good question. I would like to answer it in two aspects:

Yes, we can't distinguish between true local extinction and emigration. The observed local "extinction" of cold-adapted species over 10 years may involve two processes that usually occur in order: first "emigration" and then, if can't emigrate or withstand, "real local dieback". Over 10 years, the cold-adapted species would have to tolerate before real extinction on remote islands because of disperse limitation, while on less isolated islands it would be easy to emigrate and find a more suitable habitat for the same species. Consequently, it's harder for us to observe "extinction" of species on more isolated islands, while it's easier to observe "fake extinct" of species on less isolated islands due to emigration. As a result, the observed extinction rate is expected to increase more sharply for species on less remote islands, while the observed extinction rate is expected to increase relatively moderately for the same species on remote islands.

We have modified the legend of Figure 1 on Lines 780-781:

"Note that extinction here may include both the emigration of species and then the local extinction of species."

There is also one part in the Discussion that mentions this on Lines 287-291: "While we cannot truly distinguish in our system between local extinction and emigration, we suspect that given two islands equal except in isolation, if both lose suitability due to climate change, individuals can easily emigrate from the island nearer to the mainland, while individuals on the more isolated island would be more likely to be trapped in place until the species went locally extinct due to a lack of rescue".

Besides, you said "But what about increased local extinction, especially for small and isolated islands?", I think you are mentioning the "high extinction rate per se on remote islands". We want to test the "trend" of extinction rate on a temporal scale, rather than the extinction rate per se on a spatial scale. Even though species have a high extinction rate on remote islands, it can also show a slower changing rate in time.

I hope these answers solve the problem.

| (6) L 245: I think this is the first time the acronym appears in the ms (as the methods come after the discussion), so please write the full name here too.

Thank you for pointing out this. I realized “Thousand Island Lake” appears for the first time in the last paragraph of the Introduction part. So we add “TIL” there on Lines 108-109:

“Here, we use 10 years of bird community data in a subtropical land-bridge island system (Thousand Island Lake, TIL, China, Figure 2) during a period of consistent climatic warming.”

(7) L 319: this section could end with a summary statement on idiosyncratic responses (i.e. some variation in the responses you found among the species) and the potential reasons for this, such as e.g. the role of other species traits or interactions, as well as other ways to measure habitat fragmentation (see main comments in public review).

Thank you for this suggestion both in Public Review and here. We added a summary statement about the reasons for idiosyncratic responses on Lines 334-338:

“Overall, these idiosyncratic responses reveal several possible mechanisms in regulating species’ climate responses, including resource demands and biological interactions like competition and predation. Future studies are needed to take these factors into account to understand the complex mechanisms by which habitat loss mediates species range shifts.”

We only strengthen “habitat loss” here, because idiosyncratic responses mainly come from the mediating effect of habitat loss. For the mediating effect of isolation, the response is relatively consistent (see Page 8, Lines 183-188): “In particular, the effect of isolation on temporal dynamics of thermophilization was relatively consistent across cold- and warm-adapted species (Figure 5a, b); specifically, on islands nearer to the mainland, warm-adapted species (15 out of 15 investigated species) increased their colonization probability at a higher rate over time, while most cold-adapted species (21 out of 23 species) increased their extinction probability at a higher rate”.

(8) L 333: what about the distance to other islands? it's more of a network than a island-mainland directional system (Figure 2). You could address this aspect in the discussion.

Thank you for this good question again. Isolation can be measured in different ways in the study region. We chose distance to the mainland because it was the best predictor of colonization and extinction rate of breeding birds in the study region, and produced similar results like the other distance-based measures, including distance to the nearest landmass, distance to the nearest larger landmass (Si et al., 2014). We still agree with you that it’s necessary to consider more aspects of “isolation” at least in discussion for future research. In Discussion part, we addressed these on Lines 292-299. For more details refer to the response to Public Review.

(9) Figure 2: Is B1 one of the sampled islands? It is clearly much larger than most other islands and I think it could thus serve as an important population source for many of the adjacent smaller islands? Thus, the nearest neighbor distance to B1 could be as important in addition to the distance to the mainland?

Yes, B1 is one of the sampled islands and is also the biggest island. In previous research in our study system, we tried distance to the nearest landmass, to the nearest larger landmass and the nearest mainland, they produced similar results (For more details refer to the response to Public Review). We agree with you that the nearest neighbor distance to B1 could be a potentially important measure, but need further research. In our Discussion, we address these on Lines 292-299:

“As a caveat, we only consider the distance to the nearest mainland as a measure of fragmentation, consistent with previous work in this system (Si et al., 2014), but we acknowledge that other distance-based metrics of isolation that incorporate inter-island

connections could reveal additional insights on fragmentation effects. The spatial arrangement of islands, like the arrangement of habitat, can influence niche tracking of species (Fourcade et al., 2021). Future studies should take these metrics into account to thoroughly understand the influence of isolation and spatial arrangement of patches in mediating the effect of climate warming on species.”

| (10) L 345: 20km/h walking seems impressively fast? I assume this is a typo.

Sorry for the carelessness, it should be 2.0 km/h. it has been corrected on Lines 375-376:

“In each survey, observers walked along each transect at a constant speed (2.0 km/h) and recorded all the birds seen or heard on the survey islands.”

| (11) L 485: I had difficulties fully understanding the models that were fitted here and could not find them in the codes you provided (which were otherwise very well documented!). Could you explain this modeling step in a bit more detail?

Thank you for your recognition! According to Line 485 in the online PDF version (Methods part 4.6.3), it says: “An increasing colonization trend of warm-adapted species and increasing extinction trend of cold-adapted species are two main expected processes that cause thermophilization (Fourcade et al., 2021). To test our third prediction about the mediating effect of habitat fragmentation, we selected warm-adapted species that had an increasing trend in colonization rate (positive year effect in colonization rate) and cold-adapted species that had an increasing extinction rate (positive year effect in extinction rate)....”

We carefully checked the code in Figshare link and found that the MOSM JAGS code was not uploaded before. Very sorry for that. Now it can be found in the document [MOSM.R] at <https://figshare.com/s/7a16974114262d280ef7>. Hope the code, together with the modeling process in section 4.5 in the Methods can help to understand the whole modeling process. Besides, we would like to explain how to decide the temporal trend in colonization or extinction of each species related to Line 485. Let's take the model of species-specific extinction rate for example:

$$\text{logit}(\varepsilon_{ni,t}) = \alpha 0_n + \alpha 1_n \text{area}_i + \alpha 2_n \text{isolation}_i + (\alpha 3_n + RE_i) \text{year}_t + \alpha 4_n \text{area}_i \text{year}_t + \alpha 5_n \text{isolation}_i \text{year}_t + \text{island}_{REni}$$

In this model, “Island” was a random effect, “Year” is added as a random slope, thus allowing “year effect” (that is: the temporal trend) of extinction rate of species to vary with “island”. Further, the interaction effect between island variables (isolation, area) was added to test if the “year effect” was related to island area or isolation.

Because we are only interested in warm-adapted species that have a positive temporal trend in colonization and cold-adapted species that have a positive temporal trend in extinction, which are two main processes underlying thermophilization, we choose warm-adapted species that have a positive year-effect in colonization, and cold-adapted species that has a positive year-effect in extinction. Hope this explanation and the JAGS code can help if you are confused about this part.

Hope these explanations can make it clearer.

| (12) Figure 1: to me, it would be more intuitive to put the landscape configuration in the titles of the panels b, c, and d instead of “only” the mechanisms. E.g. they could be: a)

fragmented islands with low climate buffering; b) small islands with low habitat heterogeneity; c) isolated islands with dispersal limitations?

It is also slightly confusing that the bird communities are above "island" in the middle of the three fragmented habitats - which all look a bit different in terms of tree species and structure which makes the reader first think that it has something to do with the "new" species community. so maybe worth rethinking how to illustrate the three fragmented islands?

We would like to thank you for your nice proposition. Firstly, it's a good idea to put the landscape configuration in the title of the panels b, c, d. The new title (a) is "Fragmented islands with low climate buffering", title (b) is "Small islands with low habitat heterogeneity", and title (c) is "Isolated patches with dispersal limitations".

Second, we realized that putting the "bird community" above "island" in the middle of the three patches is a bit confusing. Actually, we wanted to show bird communities only on that one island in the middle. The other two patches are only there to represent a fragmented background. To avoid misunderstanding, we added a sentence in the legend of Figure 1 on Lines 778-780:

"The three distinct patches signify a fragmented background and the community in the middle of the three patches was selected to exhibit colonization-extinction dynamics in fragmented habitats."

(13) Figure 4: please add the description of the color code for panel a.

Sorry for the unclear description. The vertical dashed line indicates the median value of STI for 60 species, as a separation of warm-adapted species and cold-adapted species. We have added these details on Lines 807-809:

"The dotted vertical line indicates the median of STI values. Cold-adapted species are plotted in blue and warm-adapted species are plotted in orange."

(14) Figure 5: You could consider adding this as panel c to Figure 4 as it depicts the same thing as in 4a but for CTI-abundance.

Thank you for this advice. We have moved the original Figure 5 to Figure 4c. Previous Figure 6 thus turned into Figure 5. All corresponding citations in the main text were checked to adapt to the new index. The new figure is now on Lines 801-815:

References

Ferraz, G., Russell, G. J., Stouffer, P. C., Bierregaard Jr, R. O., Pimm, S. L., & Lovejoy, T. E. (2003). Rates of species loss from Amazonian forest fragments. *Proceedings of the National Academy of Sciences*, 100(24), 14069-14073. doi:10.1073/pnas.2336195100

Fourcade, Y., WallisDeVries, M. F., Kuussaari, M., van Swaay, C. A., Heliölä, J., & Öckinger, E. (2021). Habitat amount and distribution modify community dynamics under climate change. *Ecology Letters*, 24(5), 950-957. doi:10.1111/ele.13691

Gaüzère, P., Princé, K., & Devictor, V. (2017). Where do they go? The effects of topography and habitat diversity on reducing climatic debt in birds. *Global Change Biology*, 23(6), 2218-2229. doi:10.1111/gcb.13500

Gonzalez, A. (2000). Community relaxation in fragmented landscapes: the relation between species richness, area and age. *Ecology Letters*, 3(5), 441-448. doi:10.1046/j.1461-0248.2000.00171.x

Haddad, N. M., Brudvig, L. A., Clobert, J., Davies, K. F., Gonzalez, A., Holt, R. D., . . . Collins, C. D. (2015). Habitat fragmentation and its lasting impact on Earth's ecosystems. *Science advances*, 1(2), e1500052. doi:10.1126/sciadv.1500052

Richard, B., Dupouey, J. l., Corcket, E., Alard, D., Archaux, F., Aubert, M., . . . Macé, S. (2021). The climatic debt is growing in the understorey of temperate forests: Stand characteristics matter. *Global Ecology and Biogeography*, 30(7), 1474-1487. doi:10.1111/geb.13312

Si, X., Pimm, S. L., Russell, G. J., & Ding, P. (2014). Turnover of breeding bird communities on islands in an inundated lake. *Journal of Biogeography*, 41(12), 2283-2292. doi:10.1111/jbi.12379

<https://doi.org/10.7554/eLife.98056.2.sa0>